

MEDHAVA

MEDHAVA — PLAN OF ACTION

One business operating system. Any industry. One shared data core.

31 modules · 165 apps · 302 rules (89 enforced) · 151 tables · 55 product screens across 12 sectors · 19 tool capabilities · 10 industry packs. Every count in this line is read from `brand/site/modules.js`, `brand/site/rules.js`, `brand/site/shots.js`, `brand/site/tools.js`, `core/schema.postgres.sql` and `core/packs/` by `brand/site/mkcounts.js` — no figure here was typed from memory.

What this document is. The plan for building Medhava as a product that any business can adopt, from planning through execution. It is not the Vastrangam plan with the names changed. `PLAN_OF_ACTION.md` is that: one ethnic-wear group in Surat adopting this engine, with its own karigar payroll, its own three companies, its own migration off an accounting package. That document stays as it is, and it is the most useful thing in this repository — it is the worked example of the whole exercise, an entire industry implementation carried to the level of detail that proves the engine bends.

This document answers the different question: **how does the same engine serve a dental clinic and a freight forwarder and a dairy co-operative, without becoming a different program each time?**

PART I — WHAT MEDHAVA IS

M1 · ONE ENGINE, MANY TRADES

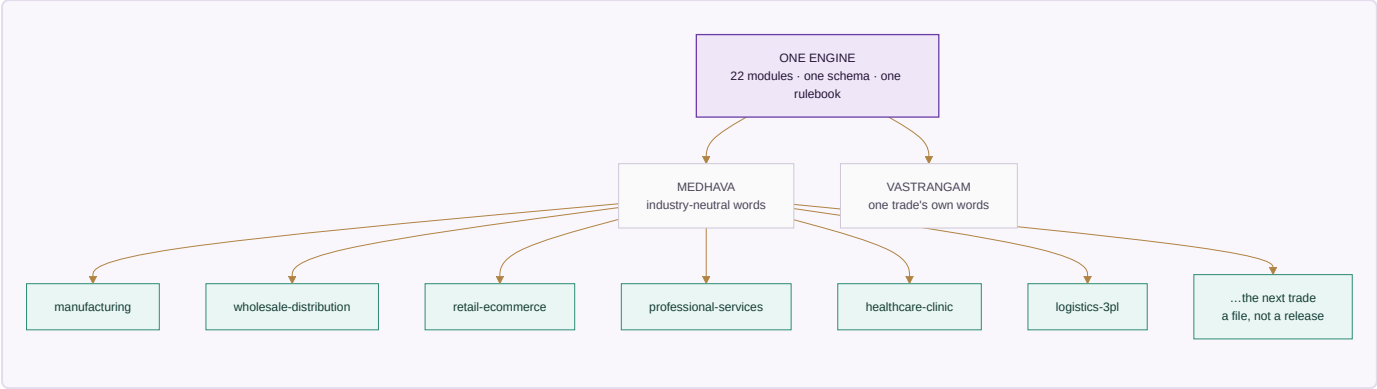
Medhava is a single application over one shared data core. A business event is entered once and flows everywhere it belongs — that law, and the cascades that follow from it, are set out in full in `PLAN_OF_ACTION.md` §A0 and are identical here, because it is the same engine. This document does not restate them.

What is specific to Medhava is the claim that the engine does not know which trade it is in. That claim is either true in the code or it is marketing, so here is where it is enforced:

The claim	What makes it true
No module assumes an industry	<code>brand/site/modules.js</code> is audited at build time; trade vocabulary in it fails the build
Trade words live in one place	The edition overlay may change wording only — <code>build.js</code> compares the structural shape before and after applying it and exits if a module number, an app name or an app count moved
The same structure ships twice already	Two editions build from one module list: MEDHAVA neutral, VASTRANGAM in one trade's own words
The number of companies is data	<code>core/tests/core.test.js</code> posts across a 10 × 10 company × channel grid, then runs 11 × 11 with no code changed

The sentence the whole product rests on. A clinic's appointment, a factory's work order, a law firm's matter, a contractor's site and a service firm's job are the same record with different words on it. Everything below is the work of making that true rather than clever.

Drawn, because the shape is the argument:



The editions differ in **wording**. The packs differ in **configuration**. Neither is a copy of the code, and that is the only reason one team can carry all of them.

M2 · THE INDUSTRY PACK ENGINE

An earlier version of this document said, in this place, that the industry pack was **specified, not built** — that a third trade meant someone writing a third overlay by hand, and that this did not scale to a product. That was the honest state of it then. It is built now, and the paragraphs below say what was built, what it refuses, and which trades it was pointed at first and why.

M2.1 · Which trades actually run this software

The pack order was not chosen by taste. It was chosen by looking up who actually buys ERP, order management and warehouse management, and building for the largest first. Published estimates differ between research houses — sometimes considerably, because "share of users" and "share of revenue" are different questions — so the figures below are attributed, and the ranking rather than any single number is what the build order rests on.

ERP — who the users are

Industry	Share of ERP users	Note
Manufacturing	~21% of users; ~32% of market revenue	The largest block on every measure. Cited elsewhere as 19.7% of revenue — the spread is why the ranking, not the decimal, is what is used
Banking, financial services, insurance	~16%	Heavily served by specialist systems rather than general ERP
Professional and financial services	13.86%	The second-largest block, and the one with no stock at all
Distribution and wholesale	9.90%	Credit and movement, not production
Healthcare	4.95%	Small today — and the fastest-growing, at 22.37% CAGR to 2030
Construction	1.98%	
Education	0.99%	

Finance and insurance, manufacturing and public administration together account for roughly 60% of total ERP spend.

WMS — who the users are

End user	Share	Note
Retail and e-commerce	~28%, about \$1.37bn in 2025, 10.1% CAGR	The largest WMS segment
Manufacturing	23.6–30.22% depending on the source	
Healthcare and pharmaceuticals	~10%	The fastest-growing, at 12.2% CAGR

The whole WMS market is put at \$3.4bn–\$5.92bn in 2025, again depending on who is counting and what they count as WMS.

OMS and third-party logistics — which markets are served

Market served by 3PLs	Share
Transportation	90%
Manufacturing	83%
Retail	83%
E-commerce	68%
Wholesale	67% (down from 83% the previous year)

And among the technology services those providers offer, **order management ranks first** — ahead of visibility (86%), transport management (84%), optimisation (77%) and ERP integration (75%). That is the finding that matters most for Medhava's shape: for a large part of this market the order, not the ledger, is the system of record.

Sources, read 2026-08-23: Grand View Research, Mordor Intelligence, HG Insights, MarketsandMarkets, Manufacturing Lead Generation, Inbound Logistics (3PL Perspectives), Electro IQ, openpr, an enterprise suite. These are market-research estimates, not measurements; they are used here to order a build queue, which is what they are good for, and not quoted anywhere in the product as fact.

M2.2 · What a pack is

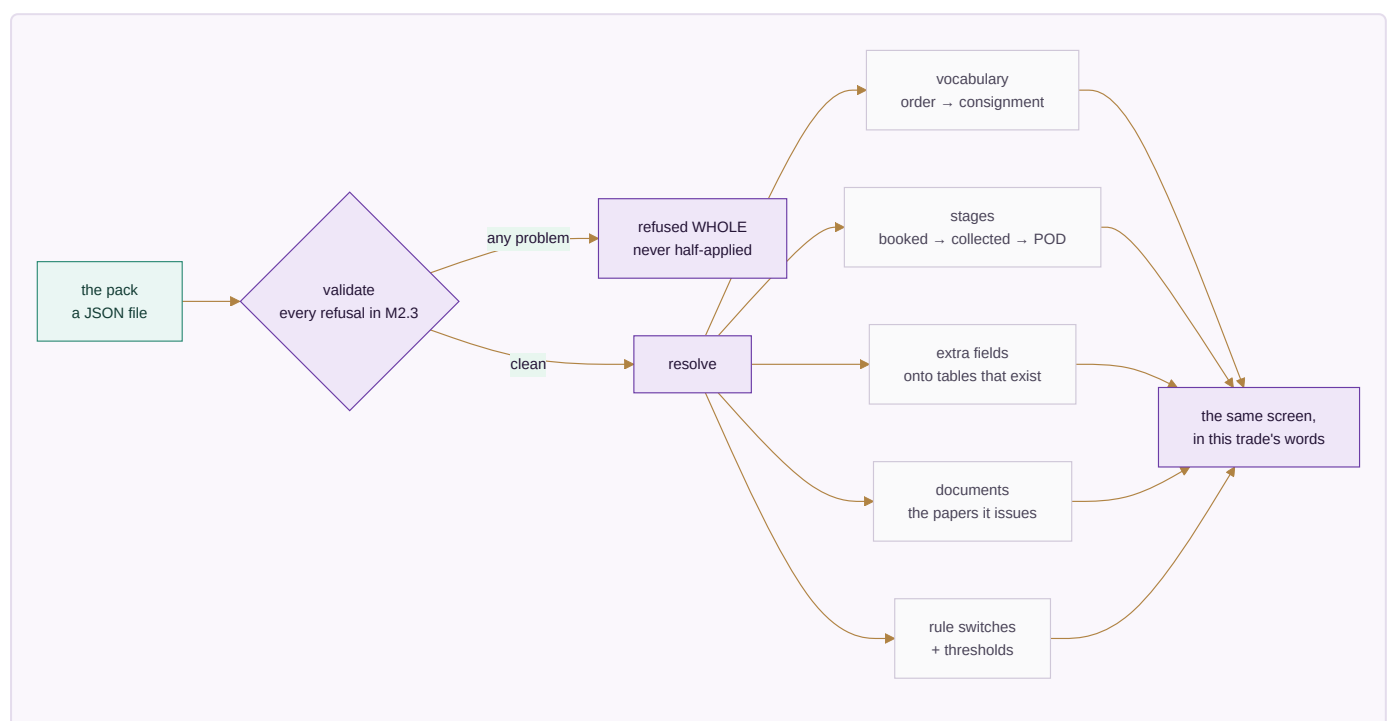
An industry pack is a row of configuration, loaded as data, that carries:

- **vocabulary** — what this trade calls an order, a customer, a unit of work, a stage
- **stages** — the pipeline a job moves through, named and ordered by the trade
- **fields** — the extra attributes this trade needs on a record, and their types
- **documents** — the papers it issues, and what has to be on them
- **rule switches** — which discretionary rules apply, and at what thresholds
- **starting data** — a chart of accounts, roles, units of measure

The engine reads a pack the way it already reads companies and channels: as rows. Nothing in `modules.js`, `core/` or the schema changes when a fourteenth trade is added — which is exactly the property the 10×10 test already proves for companies, applied to trades.

M2.2a · How a pack becomes a screen

Nothing about this is magic, and the picture is the fastest way to see that:



M2.3 · What a pack may never do

A configuration file that can do anything is not configuration, it is a hole. Six refusals are enforced in `core/packs.js` and each one is a named test in `core/tests/packs.test.js`:

A pack may not	Because
contain executable code, at any depth	the moment a pack can run code, adding a trade is a code change again and the whole guarantee is worthless
invent a concept the engine does not have	"vocabulary" would quietly become a place to put anything, and the screens would carry a name for something that does not exist
add a field to a table that does not exist	the shape of the database would move outside the reach of the schema test that guards it
declare money as a plain number	the entire schema was built to keep floating-point rupees out; a pack is not a side door
switch off an immutable rule	a trade may call an invoice a fee note; it may not decide its audit trail is optional
be applied in part	a half-loaded trade is a system whose vocabulary and rules disagree, with no way to tell which half is live

The rule ids marked immutable in `core/packs.js` cover company scoping, the audit trail, the posting rules, group elimination and roster privacy — and a test asserts that every one of them really exists in the rulebook, so the protection cannot silently guard nothing.

One default is worth stating on its own, because getting it backwards would be invisible: **a rule a pack never mentions is ON**. The rulebook is the default and a pack is an exception list, never a permission list. The other way round, every rule added after a pack was written would silently apply to nobody using it.

M2.4 · Which packs ship

Built in the order the research ranks them:

Rank	Pack	Sector	What it is really for
1	manufacturing	Manufacturing	The largest ERP user base, and the vocabulary furthest from a service firm's
2	wholesale-distribution	Distribution	Credit, not production, is the thing the software has to hold
3	retail-ecommerce	Retail	The largest WMS segment; returns and settlement are the hard part
4	professional-services	Services	No stock at all — the hardest case for the neutrality claim
5	healthcare-clinic	Healthcare	Small now, fastest-growing on every measure
6	logistics-3pl	Logistics	The most-served 3PL market, where the order <i>is</i> the product

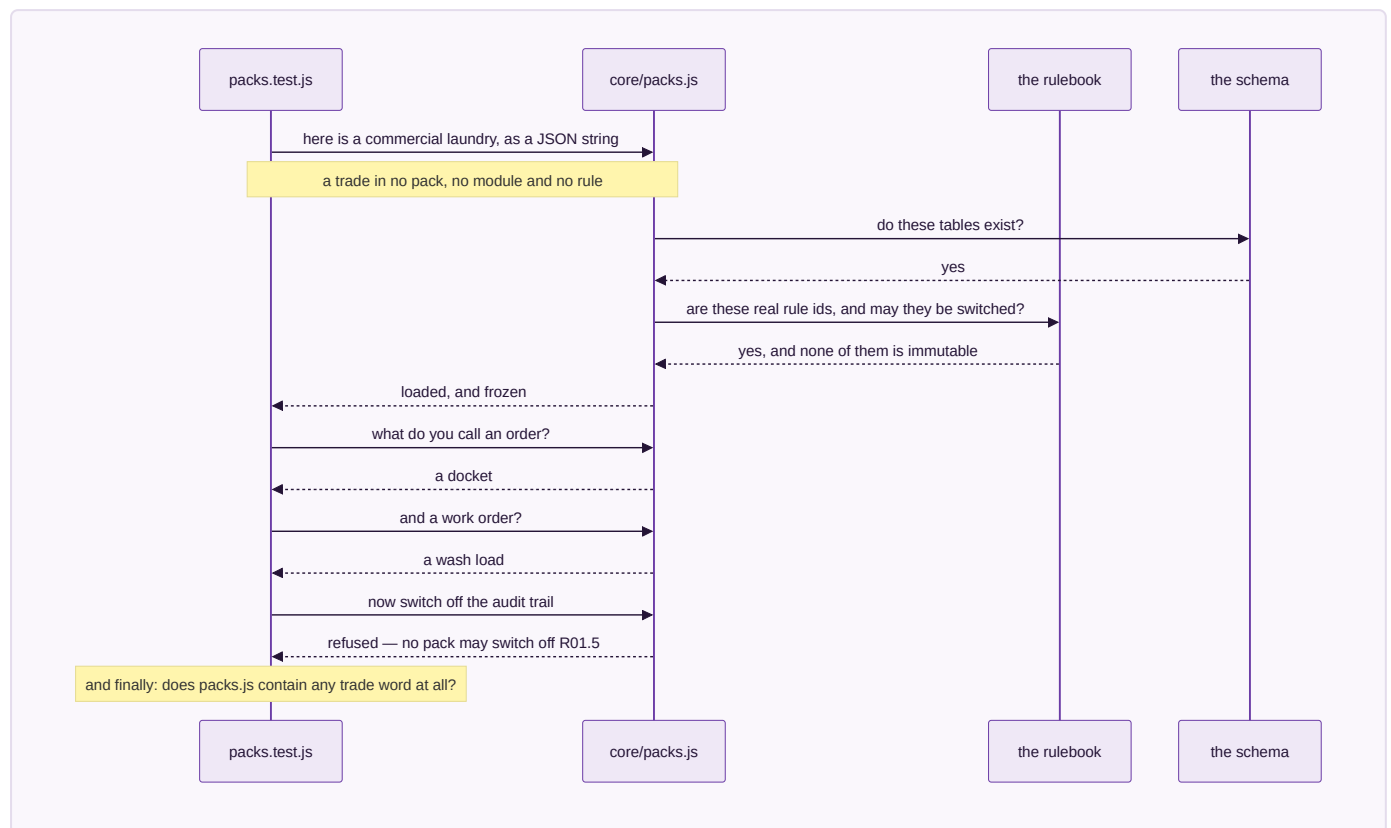
M2.5 · The gate, and what it proves

Phase 2's gate was written before the engine was: *a third trade is added without writing code.*

`core/tests/packs.test.js` §4 runs it, and runs it harder than the phase asked. During the test run it invents a **commercial laundry** — a trade that appears nowhere in this repository, in no pack, in no module, in no rule — hands the engine a JSON string, and then requires the whole system to answer in that trade's words: an order reads as a docket, a work order as a wash load, a customer as an account. Its pipeline resolves ordered and terminating, its extra fields land on real tables, its rule switches resolve against the real rulebook, and it is refused the audit trail exactly as the six shipped packs are.

A last assertion closes the loop: `core/packs.js` **may not contain a single trade word.** Not laundry, not clinic, not freight, not dealer, not matter, not patient. If the engine ever learns a trade, the test fails — because an engine that knows one trade's words has an opinion about which trades are normal, and that is the failure this whole section exists to prevent.

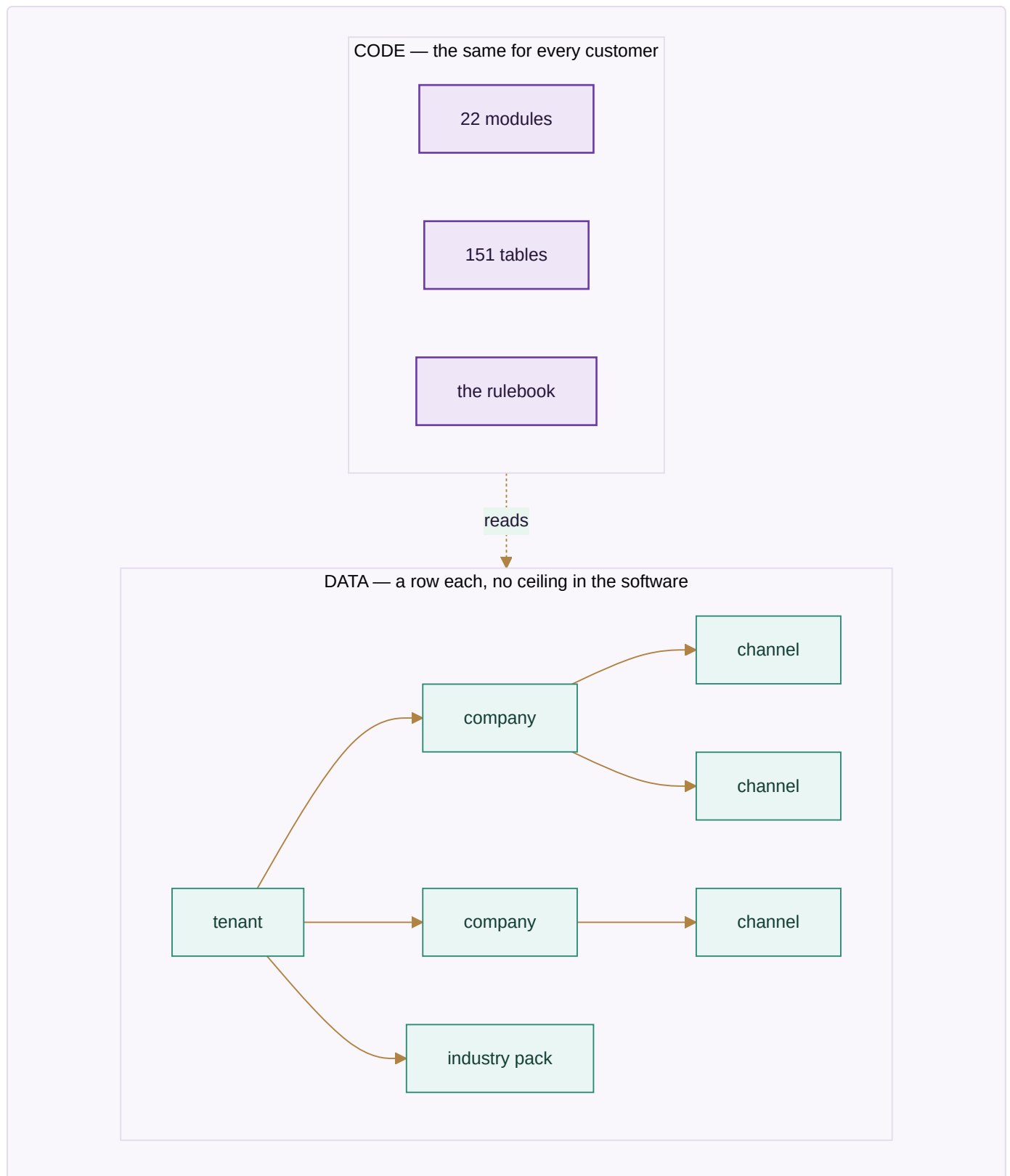
What the gate actually does, step by step:



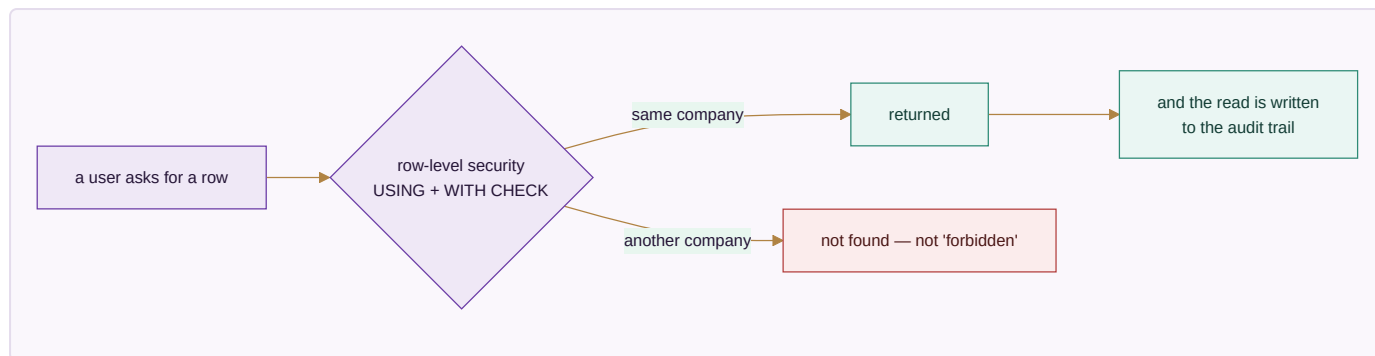
`node core/tests/packs.test.js` → **every check passes, 0 failures.**

M3 · TENANCY — WHAT IS A ROW AND WHAT IS CODE

Concept	Row or code	Consequence
Tenant (a customer of Medhava)	row	Onboarding a business is data entry, not a deployment
Company inside a tenant	row	A group with four companies is four rows, consolidated with inter-company trade eliminated
Channel	row	A new marketplace is a row; rule R15.1 makes discovery automatic
Location, stage, role	row	A warehouse, a production stage and a job title are all configuration
Industry pack	row	A trade is configuration, not a fork — <code>core/packs.js</code> , six packs shipped, a seventh added during the test run
The 22 modules	code	The structure is the product; it is the same for everyone
The rulebook	code	Which apply is configurable; what they refuse is not — and 22 of them cannot be switched off by any pack



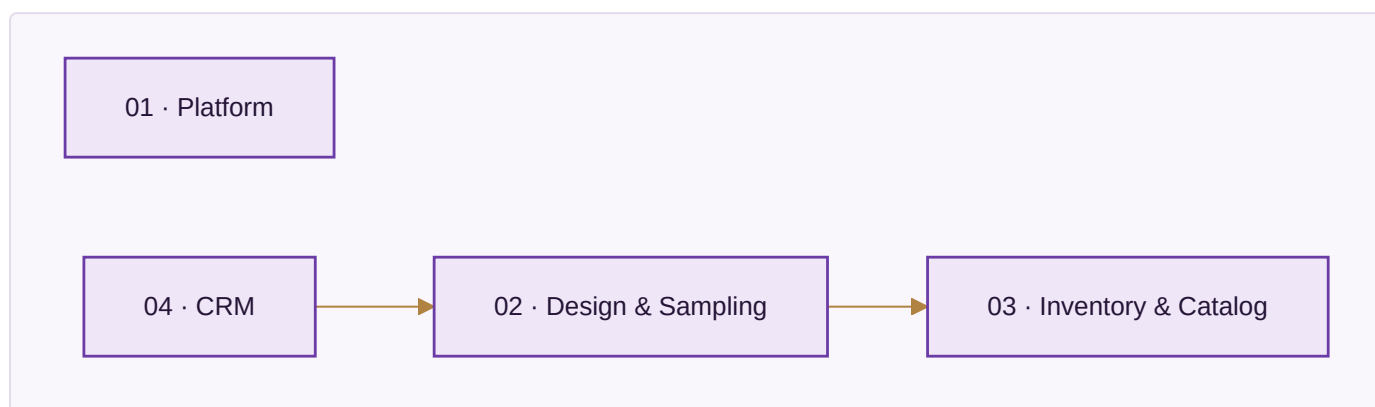
Isolation. Every business table carries `company_id` and a row-level security policy carrying both `USING` and `WITH CHECK`, so a read and a write are separately prevented from crossing a boundary. `core/tests/schema.test.js` fails the build if any company-scoped table lacks one. Cross-tenant isolation is the same mechanism one level up and is the single highest-risk item in this plan — a bug there is not a defect, it is an incident.



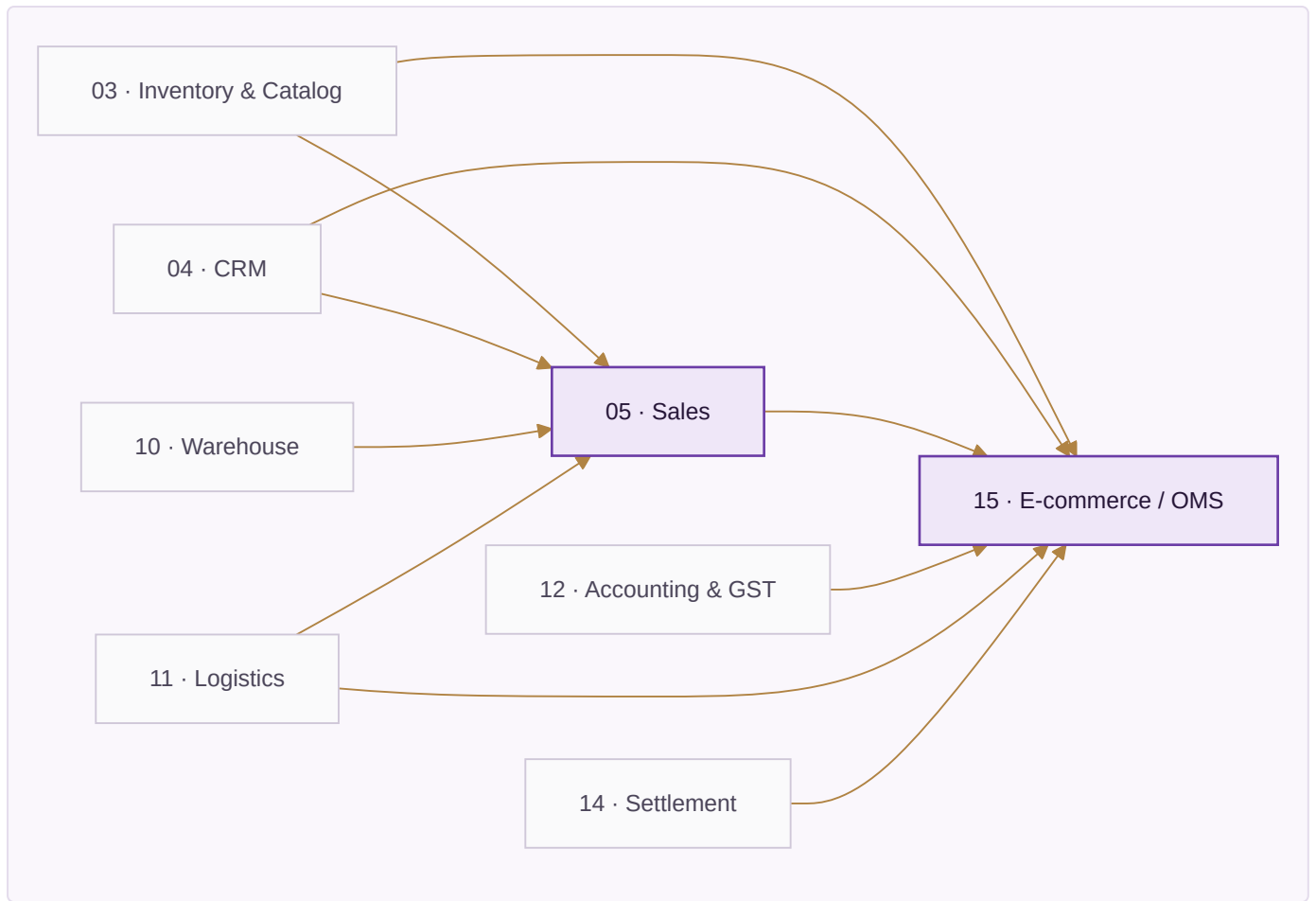
M4 · THE 22 MODULES, READ FROM TWELVE TRADES

How the modules feed each other. Generated from the `reads` field on every module in `brand/site/modules.js` by `brand/site/mkdiagrams.js` — the same field the website renders as "Reads from" — so it cannot drift from the module list. Cut into bands because all 22 modules and all 44 edges on one page rendered as an unreadable tangle; the information is the same, the page is legible.

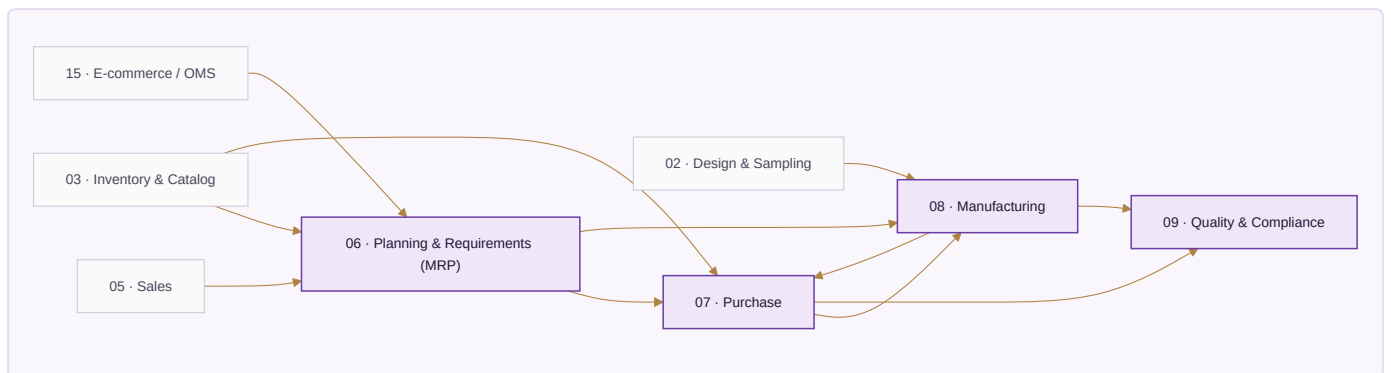
Foundation — what it reads, and from where.



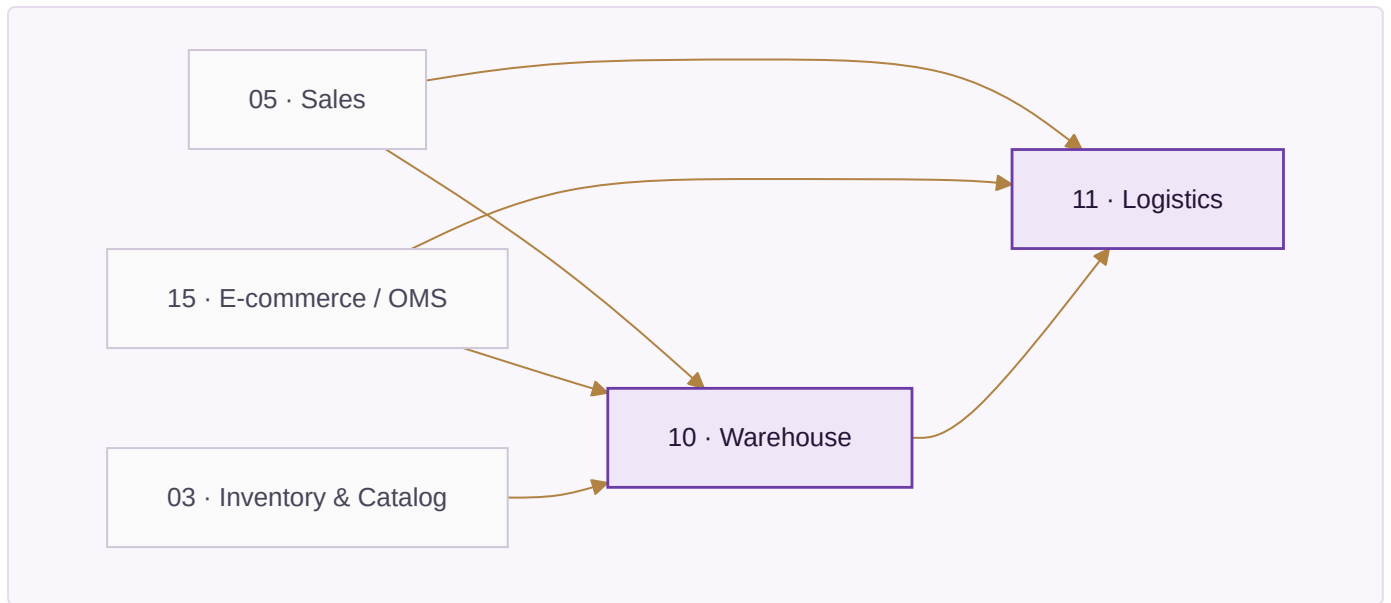
Selling — what it reads, and from where.



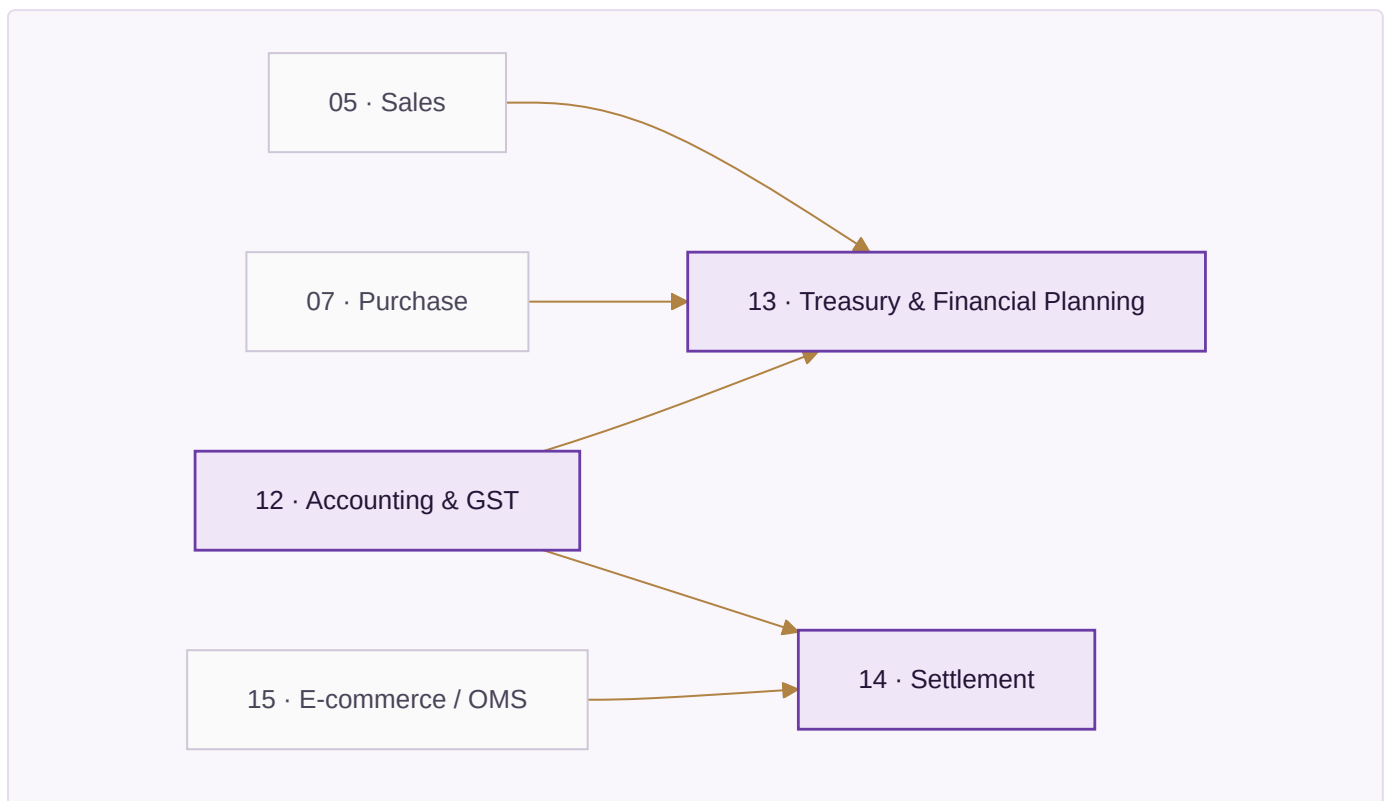
Planning & making — what it reads, and from where.



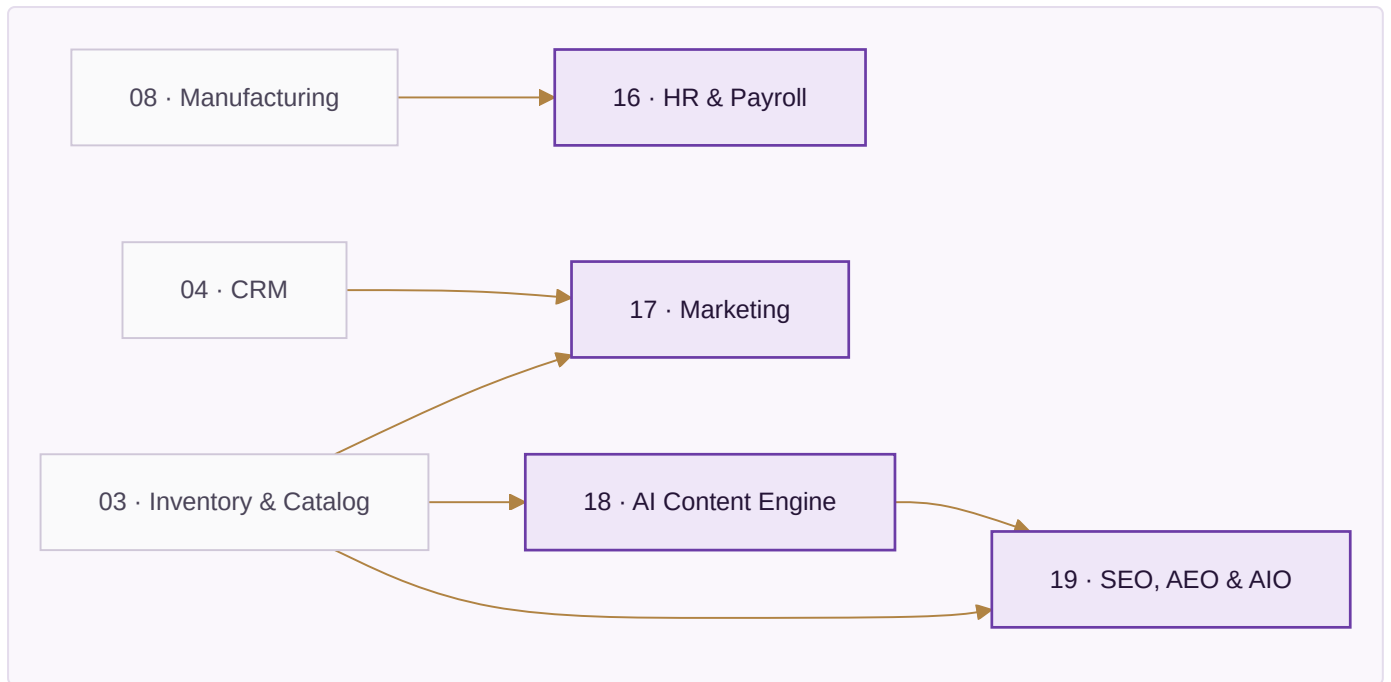
Moving it — what it reads, and from where.



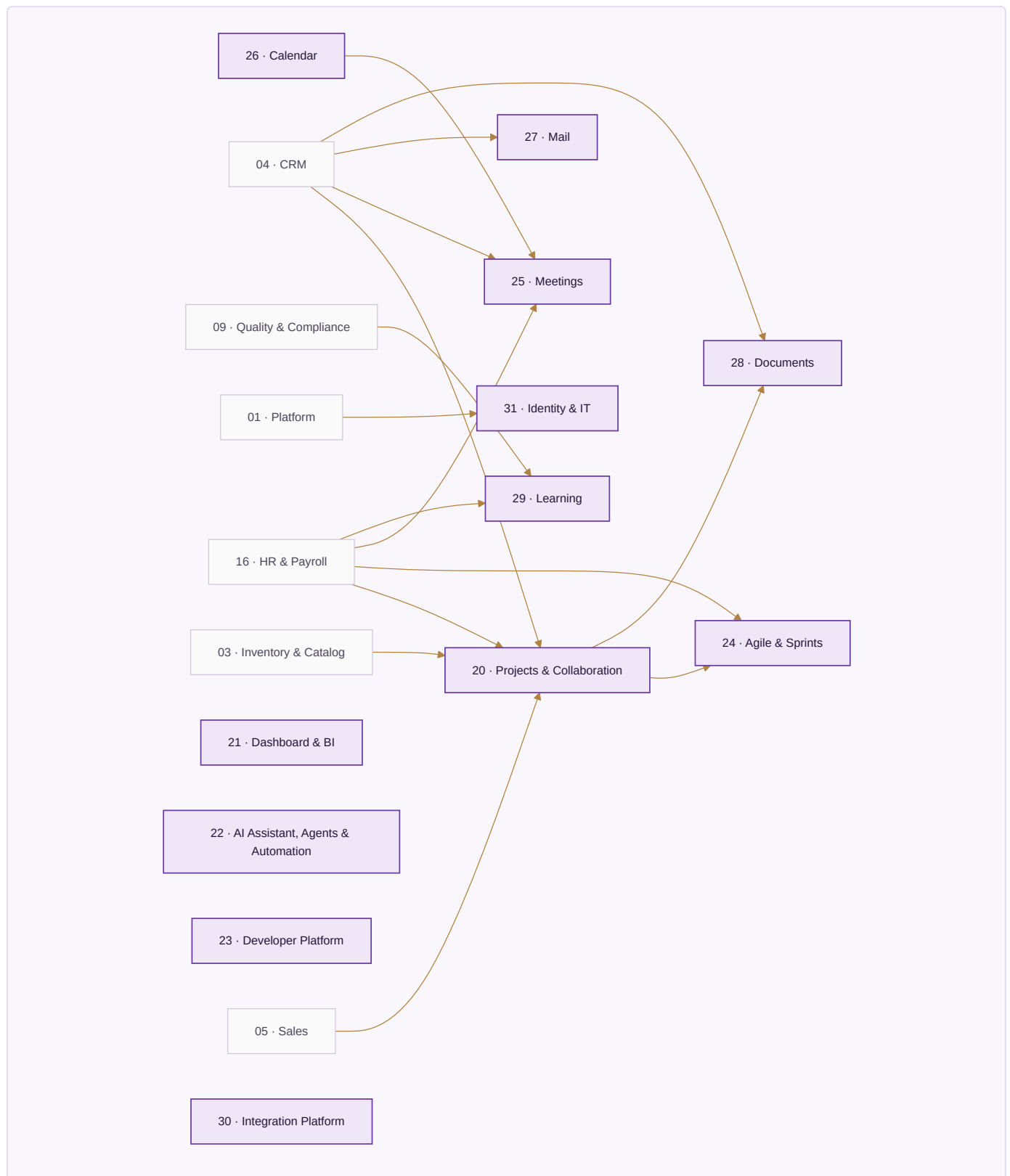
The money — what it reads, and from where.



People & demand — what it reads, and from where.



Across the business — what it reads, and from where.



01, 03, 04, 12, 21, 22, 23, 26, 27, 30, 31 declare that they read *every module*: they sit on the shared data core rather than on any one upstream module, which is why no arrow into them is drawn above. Everything else reads exactly what the arrows show.

The module list is in [PLAN_OF_ACTION.md](#) Part II in full, with every app and every rule. It is not repeated here. What is here is the reading that makes it industry-neutral — the same module, seen from a different trade:

Module	Manufacturing	Professional services	Healthcare	Hospitality
02 Design & Sampling	part revision	matter scoping	protocol	recipe development
05 Sales	order book	engagement letter	appointment book	cover forecast
06 Planning	material requirement	resourcing	rota and stock	purchase plan
08 Manufacturing	work order, stages	—	—	batch and prep
09 Quality	inspection	file review	licence and calibration	food safety log
10 Warehouse	pick and pack	—	consumables	store issue
16 HR	shift and piece-rate	timesheets, chargeable	sessions and locums	rota and casuals
20 Projects	improvement work	matters	care pathways	openings

The landing page carries every one of the product screens counted at the head of this document, across all twelve sectors, doing exactly this — the same module rendered with a machine shop's figures and a clinic's figures, one under the other. That is the argument made where it can be checked rather than believed.

PART II — EXECUTION

M5 · THE STACK, AND WHAT IT COSTS

The full tools register is `brand/site/tools.js`, gated by `brand/site/checktools.js`, which fails the build if any paid tool does not name **both** the free option it replaces and the concrete condition that forces the upgrade. "When we grow" is rejected by the checker; a number or a named event passes.

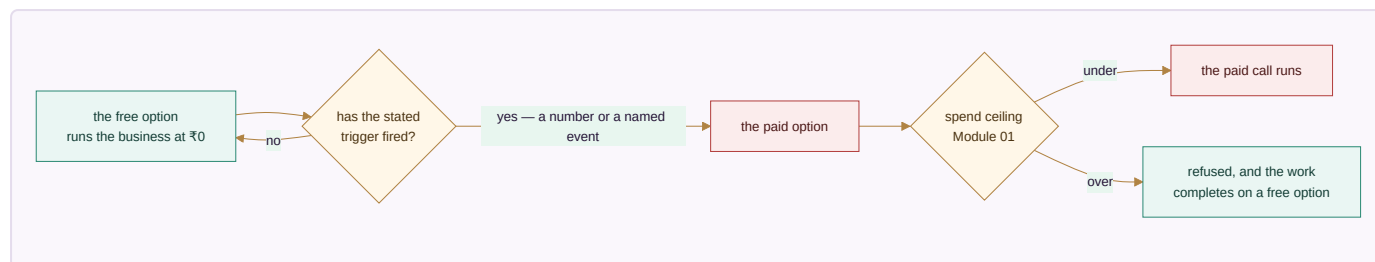
19 capabilities. Only three have no free path in existence:

Capability	Why there is no free option	What it costs
WhatsApp Business messaging	Meta requires an approved provider; there is no free tier to outgrow	~₹1,500–3,000/month plus per-conversation charges
SMS	Carrier access is the cost	~₹0.15–0.25 per message
Generated imagery	Free software, but it needs a graphics card	A GPU, or a hosted API per image

Everything else runs on free tiers or free software until a stated trigger fires: Postgres, self-hosted automation, a free hosting tier, local AI, the browser's own speech synthesis, ffmpeg and Chromium for media, UPI for payments, CSV for every integration, and a progressive web app instead of a store listing. **A business can run the whole system on zero rupees of software licence** and pay only for the three lines above, and only when it reaches them.

The Provider Router in Module 01 puts a spend ceiling in front of every paid call and refuses past it rather than warning — so the AI line in particular cannot quietly become the largest one.

How a line ever becomes a paid line. Every capability starts free and only moves when a stated condition fires — which is the whole discipline `brand/site/checktools.js` exists to enforce:



A paid entry with no free predecessor and no concrete trigger **fails the build**. "When we grow" is not a trigger; a number is.

M6 · THE EIGHT PHASES

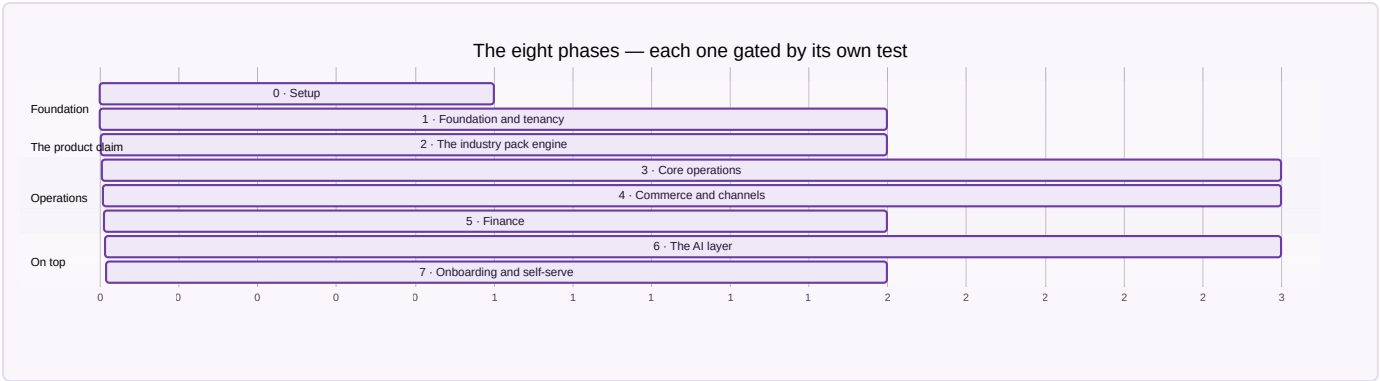
The gate is absolute: **Phase N+1 does not start until Phase N's tests pass**. A phase is done when its stated result is reproduced, not when its code is written.

Phase	What gets built	Done when
0 · Setup	Environments, CI, monitoring, the schema loaded	A commit deploys and a user logs in
1 · Foundation & tenancy	Tenants, companies, roles, RLS, masters, the SKU/item model	Two tenants exist and neither can read a single row of the other, proved by a test that tries
2 · The industry pack engine	Vocabulary, stages, fields, documents and starting data as rows	A trade nobody designed for is added without writing code , during the test run, from a plain settings file
3 · Core operations	Inventory, procurement, production, quality, warehouse	Three trades run the same operations screens on their own vocabulary
4 · Commerce & channels	Sales all channels, OMS, returns, logistics, settlement	A week of orders across channels, settled and reconciled
5 · Finance	Double-entry, tax, banking, period locks	A month closes: trial balance ties and returns generate from vouchers
6 · The AI layer	Assistant, chatbot, agents, guardrails, content	An assistant answer matches the books; an agent asked to move money stops
7 · Onboarding & self-serve	Sign-up, pack selection, import, go-live	A business onboards itself in a day without a call

Phase 2 is the one that decides whether this is a product or a project. Its gate is written before its engine — *a new trade must be addable without a developer* — and it runs on every test pass against a trade nobody designed

for. Everything from Phase 3 onward stands on that claim, so it is checked rather than argued.

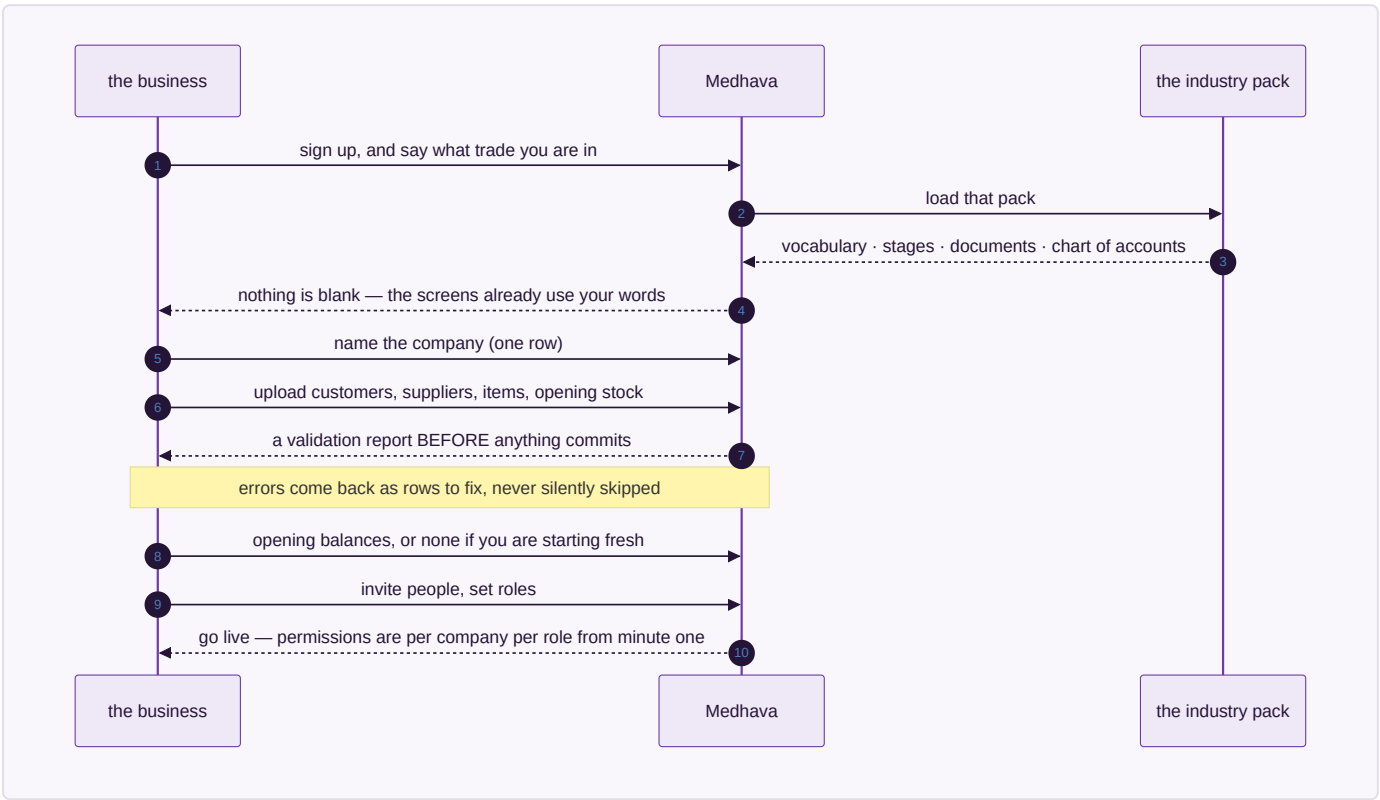
The same eight phases as a sequence:



Bars show sequence and relative size, not calendar dates — a phase ends when its test passes, and a date typed here would be a promise the gate does not make.

M7 · ONBOARDING A BUSINESS IN A DAY

For a business with no system to migrate from — which is most of the target market — the sixty-day parallel run in the Vastrangam plan is the wrong shape entirely. The sequence is:



1. **Sign up, pick a trade.** The industry pack loads vocabulary, stages, documents and a chart of accounts. Nothing is blank.
2. **Name the company.** One row. A group adds more rows now or later.

3. **Import what exists** — customers, suppliers, items, opening stock — from a spreadsheet, with a validation report **before** anything commits. Errors are shown as rows to fix, never silently skipped.
4. **Opening balances**, or none at all if the business is starting fresh.
5. **Invite people, set roles**. Permissions are per company per role from the first minute.
6. **Do one real transaction end to end** — one sale, invoiced, stock moved, posted — and click the dashboard figure down to the voucher. That is the go-live test, and it takes a minute.

For a business migrating from an existing system, the discipline in `PLAN_OF_ACTION.md` Part IV E5 applies unchanged: export, clean, load, opening balances, parallel run, and a decommission criterion agreed in advance rather than argued at the end.

M8 · SECURITY AND COMPLIANCE

Row-level security in the database and permission checks in API middleware — one layer is one mistake away from a cross-tenant read. Personal data is exportable and erasable on request, with consent and retention tracked as the two separate clocks they are. Card data never reaches this system; the gateway's own secured field takes it, so there is no scope to protect. The audit trail has no off switch, and R16.22 keeps individual pay and personal attributes out of any document that leaves the building.

M9 · PERFORMANCE

Operation	p95
Page load, cached / uncached	< 1 s / < 3 s
Live stock or availability query	< 500 ms
Document PDF	< 2 s
Channel order pull, per channel	< 60 s
Settlement import, 1,000 lines	< 30 s
Daily profit-and-loss refresh	< 10 s

The availability query is the one that decides whether people trust the system: it runs on every screen showing a quantity or a free slot, and one that takes two seconds is one people stop asking.

M10 · RISK REGISTER

Risk	Likelihood	Impact	Mitigation
Cross-tenant data leak	Low	Critical	RLS with USING and WITH CHECK, middleware checks, a test that actively attempts a cross-tenant read
The industry pack engine slips, and trades get hard-coded instead	High	Critical	Phase 2 gate: a third trade added with no code. Hold the gate
A vertical demands a genuine structural exception	Medium	High	Add it as a module or an app for everyone, never as a branch for one customer
Free tier terms change or a tier disappears	Medium	Medium	Every capability has a self-host route recorded; the register is dated
WhatsApp provider outage	Low	High	Adapter swap; SMS and in-app as fallback
Onboarding needs hand-holding, so it does not scale	High	Medium	The day-one sequence above is a tested path, not a document
AI spend runs away	Medium	Medium	The spend ceiling refuses rather than warns

M11 · SUCCESS METRICS

Measure	Target
A new trade supported without writing code	Yes, from Phase 2 onward — binary, not a percentage
Time for a business to onboard itself	Under a day, unattended
Cost to run for a business at pilot scale	₹0 in software licence
Rules enforced by a test	the enforced-of-total figure at the head of this document; up every build
Cross-tenant incidents	Zero, and this is the only metric with no acceptable non-zero value

M12 · RUNBOOKS

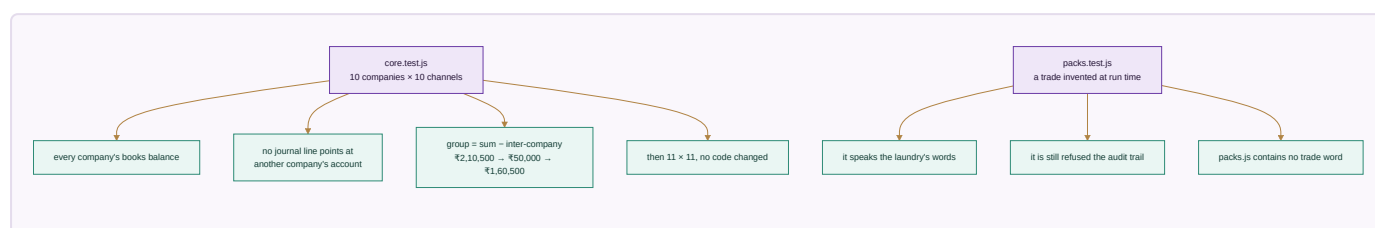
Daily — error and uptime check, failed integrations queue, onboarding funnel. **Weekly** — usage per tenant, slow queries, support themes that suggest a missing rule. **Monthly** — free-tier headroom against triggers, spend against ceilings, backup restore test. **Quarterly** — re-read the tools register against current published tiers, review which SPECIFIED rules became enforceable, and check whether any vertical has quietly grown an exception.

M13 · WHAT A CUSTOMER PAYS FOR

The free-first register is not only how Medhava is built — it is the shape of what it can offer. Because the system runs on free tiers and free software at pilot scale, a business can be given a genuinely working system before it pays anything, and the paid lines are the ones it would have had to pay anyway: messaging, marketplace access, payment processing, courier pickup.

The commercial decision — subscription, per-company, per-user, or usage — is not made in this document, because it is a business decision rather than an engineering one. What this document fixes is the constraint it must respect: **nothing in the architecture may make a plan cap technically necessary**. Companies, channels, users and trades are rows. If a plan limits them, that is a commercial choice stated in the pricing, and the software says so rather than pretending to a technical limit it does not have.

PART III — THE PROOF



The test of whether this is one system or twenty-two programs sharing a login is a single transaction followed end to end, and it is re-run at every module boundary. That walkthrough is in `PLAN_OF_ACTION.md` Part V and is identical here.

The test of whether it is *industry-neutral* is different, and it is this: **take the same transaction and run it in a trade nobody had in mind when the code was written**. That test used to be run by hand, by writing an overlay. It is now run by the machine: `core/tests/packs.test.js` invents a commercial laundry every time it executes, loads it from a JSON string, and requires the system to answer in that trade's words — while still refusing it the audit trail. The last assertion in that file is that `core/packs.js` contains no trade word at all, so the engine cannot quietly learn one trade and call it neutrality.

Every check passes, with the shipped packs and a seventh added at run time. That is the whole industry-neutrality claim, reduced to something that either passes or does not.

M14 · THE RULEBOOK

The specification a developer implements, and the promise a customer relies on. Counted in every other section of this plan; printed here in full, because a count tells a reader how much they are not being shown.

302 rules. Every one states what happens **and what the system will never do instead**. The second half is the part worth reading — it is what you are relying on when nobody is looking.

Module 01 · Platform — 33 rules

R01.1 Every business record names the company it belongs to

- **When** any record is written — a sale, a movement, a voucher, an employee
- **Then** its company is stored on the row itself
- **Never** inferring the company from who happens to be logged in, which silently mis-files every record made by someone who works across two of them

R01.2 One company cannot read another company's records

- **When** a query runs for a user scoped to one company
- **Then** rows belonging to any other company are not returned at all
- **Never** filtering in the screen while the data is already loaded — a filter can be removed, a scope cannot

R01.3 The audit trail has no off switch

- **When** anything touching money, stock, price, tax, pay or master data changes
- **Then** the change and its before-image are written in the same transaction as the change itself
- **Never** allowing a setting, a role or a migration to disable it — either both land or neither does

R01.4 An update records what it was, not only what it became

- **When** a row is changed
- **Then** the current row is read first so the before-image is what was really there
- **Never** trusting the caller's idea of the old value, which makes the trail a record of intentions rather than of facts

R01.5 A table nobody thought to audit is refused

- **When** code writes to a table that is not on the audited list
- **Then** the write is refused and the table is named
- **Never** letting it through quietly, which is how a money column ends up outside the trail without anyone deciding that

R01.6 Deletion is a reversal, never a removal

- **When** a user deletes anything
- **Then** the record is voided, marked, and still readable with the reason
- **Never** removing the row — eight years of trail cannot survive a DELETE

R01.7 A module that is not in the canonical list cannot join the bus

- **When** code subscribes to a business event
- **Then** the module number is checked against modules.js and refused if unknown
- **Never** letting an unregistered listener attach, which is how a cascade gains a step nobody can find later

R01.8 A cascade is all of it or none of it

- **When** one business event fans out to stock, ledger, customer and documents
- **Then** every step commits together, or the whole thing is rolled back
- **Never** leaving stock moved and the ledger unposted, which is the exact state no report can ever explain

R01.9 A handler that throws takes the transaction with it

- **When** any subscriber to an event fails
- **Then** the emitting transaction fails too
- **Never** swallowing the error so the originating action appears to have succeeded

R01.10 No capability depends on a single outside service

- **When** any capability is used — books, courier, payments, AI, storage, GST
- **Then** an ordered list of interchangeable providers is tried, ending on one that needs nothing connected
- **Never** having one provider whose outage stops the work, however good that provider is

R01.11 A failing provider is taken out of the list, not hammered

- **When** a provider fails repeatedly
- **Then** it is tripped open, skipped entirely, and retried once after a cooldown
- **Never** retrying into a dead service on every call while a working alternative sits further down the list

R01.12 A spend ceiling refuses, it does not warn

- **When** a paid call would take spending past the ceiling set for it
- **Then** that provider is refused and the work completes on a free one
- **Never** letting it through with a warning nobody reads, and never refusing only the first provider while the same spend reroutes to the next

R01.13 The system never asks for a marketplace, bank or account password

- **When** any integration is connected, by any module, including a chatbot or an agent
- **Then** a scoped, revocable key is requested instead, cancellable from the provider's side without changing the login
- **Never** accepting, storing, echoing or transmitting an account password — there is no screen, no import and no support flow that takes one

R01.14 Card and bank credentials never reach application code

- **When** a payment needs a card or bank detail
- **Then** the provider's own secured field takes it directly
- **Never** passing it through this system, even in transit, even unlogged — what is never received cannot be leaked

R01.15 Consent and retention are two different clocks

- **When** a person's data is held

- **Then** why it may be used and how long it is kept are tracked separately, and an erasure request is resolved against both
- **Never** treating a legal retention period as consent to keep using the data for anything else

R01.16 A scoped key is revocable without touching the login

- **When** an outside service is connected
- **Then** a key limited to what that capability needs is stored, and the connection records which capability it serves
- **Never** storing a credential that can do more than the capability requires, because the day it leaks is the day that difference matters

R01.17 A webhook is verified, idempotent and never silently dropped

- **When** a payment, courier, storefront or messaging provider calls in
- **Then** the signature is checked, the external id makes a repeat delivery a no-op, and a failure is logged with its payload for retry
- **Never** trusting an unsigned call, and never processing the same external id twice — a duplicated payout or a duplicated order is indistinguishable from a real one afterwards

R01.18 A trade is added as data, never as a version of the software

- **When** a business in a trade the system has never seen signs up
- **Then** its vocabulary, stages, extra fields, documents, rule switches and starting reference data arrive as one configuration file, and every screen reads back in that trade's words
- **Never** a branch, a fork or a bespoke build per industry — that is a consultancy with software attached, and it is the thing that stops a product from being one

R01.19 A pack is data and can never be code

- **When** a pack is loaded from any source
- **Then** every value in it is inspected, at any depth, and a function anywhere inside it refuses the whole pack
- **Never** letting configuration carry behaviour — the moment a pack can run code, adding a trade is a code change again and the guarantee in R01.18 is worthless

R01.20 A pack may rename a concept, never invent one

- **When** a pack declares its vocabulary
- **Then** each entry is matched against the fixed list of concepts the engine has, and an unknown one refuses the pack
- **Never** accepting an unrecognised word as a new concept, which turns "vocabulary" into a place to put anything and leaves the screens with a name for something that does not exist

R01.21 A pack extends tables that exist, and nothing else

- **When** a pack adds fields
- **Then** the table is checked against the real schema and the field type against the types the engine can store

- **Never** creating a table on a customer's behalf from a configuration file, which puts the shape of the database outside the reach of the schema test that guards it

R01.22 Money in a pack is money everywhere else

- **When** a pack adds a field whose name reads as an amount, a price, a cost, a total, a fee or a rate
- **Then** it must be declared in paise, and a plain number refuses the pack
- **Never** letting a trade introduce a floating-point rupee through the side door after the whole schema was built to keep them out

R01.23 No pack can switch off a guarantee

- **When** a pack sets a rule off
- **Then** the rule id is checked against the rulebook, and against the list of rules no pack may touch — company scoping, the audit trail, the posting rules, group elimination and roster privacy
- **Never** a trade opting out of the things that make the books trustworthy; it may call an invoice whatever it likes and may not decide its trail is optional

R01.24 A rule a pack never mentions is on

- **When** a rule is looked up for a trade
- **Then** the rulebook is the default and the pack is read as an exception list — silence means the rule applies
- **Never** treating the pack as a permission list, which would mean every rule added after a pack was written silently applies to nobody who is using it

R01.25 An invalid pack is refused whole, never half-loaded

- **When** a pack fails any check
- **Then** every problem in it is reported at once and none of it is applied
- **Never** partially loading a trade, which leaves a system whose vocabulary and rules disagree with each other and no way to tell which half is live

R01.26 A backup is not a backup until it has been restored

- **When** a backup is taken on its schedule
- **Then** it is restored into a scratch database on a stated interval and the row counts of the restored copy are compared against the source
- **Never** counting a backup as protection because the job exited zero, which is how a business discovers on the worst day of its life that it has been writing an empty file nightly

R01.27 What is watched is written down, with the number that means trouble

- **When** the system runs in production
- **Then** each watched signal names the level that counts as trouble and the person it wakes — disk left, error rate, queue depth, response time, failed sign-ins, and the age of the last successful backup
- **Never** a dashboard nobody is on the hook for, which is a screen that gets looked at after somebody has already noticed by other means

R01.28 An alert names what to do, not just what is wrong

- **When** a watched signal passes the level that counts as trouble and somebody is woken
- **Then** the alert carries the signal, the value, the level it passed, and the first step to take
- **Never** paging a person with a metric and no instruction, which turns every incident into a research project starting from zero at three in the morning

R01.29 A change records what it was, as well as what it became

- **When** a business record is created or changed
- **Then** the audit row carries the previous value, the new value, who made the change and when
- **Never** logging only the new value, which tells you the number is wrong today and nothing about what it was before somebody changed it

R01.33 The audit trail is kept for a period somebody chose

- **When** audit rows age past the retention period
- **Then** the period is read from configuration, and anything removed is removed on that stated rule
- **Never** a retention that exists only in whoever set up the log rotation, which is how the one month somebody needs is the one month that was rotated away

R01.30 Nothing goes live until the old numbers and the new ones agree

- **When** data is migrated from whatever the business ran before
- **Then** the migration reconciles: totals, row counts and opening balances are compared against the source and every difference is explained before cutover
- **Never** going live on a migration whose differences were noticed and waved through, because every one of them becomes a figure somebody has to defend later with no record of where it came from

R01.31 The old way keeps running until the new one has agreed with it

- **When** a module is ready to go live
- **Then** both run over the same period and their outputs are compared, and the old one is retired only after they agree
- **Never** a cutover on a date rather than on evidence, which makes the first real disagreement a production incident instead of a finding

R01.32 A rollback is rehearsed before it is needed

- **When** a release is prepared
- **Then** the way back is written down and practised on a copy, including what happens to records created after the release
- **Never** a rollback plan that was written and never run, which is a plan in the same sense as an untested backup — believed until the one moment it has to work

Module 02 · Design & Sampling — 7 rules

R02.1 A style becomes a SKU only after sign-off

- **When** someone tries to create a catalogue record for a design
- **Then** the design must already have passed sample sign-off
- **Never** letting a SKU exist for something with no agreed specification, which puts an unmakeable item on sale

R02.2 Every version of a specification is kept

- **When** a sample round changes a measurement, a fabric or a trim
- **Then** a new version is written and the old one stays readable
- **Never** editing the specification in place — a worker paid against last month's spec must still be able to show what it said

R02.3 A costed trial carries the date its rates came from

- **When** a sample is costed
- **Then** the rate and the date it was in force are both stored on the trial
- **Never** recosting an old trial with today's rates and presenting the result as what it cost then

R02.4 A design with no ownership record is flagged, not blocked

- **When** a design reaches sign-off with no trademark or copyright status on file
- **Then** it proceeds and is listed as unprotected
- **Never** silently treating it as protected, which is only discovered when a near-identical listing appears and there is nothing to act on

R02.5 The first-shown date is recorded when it happens

- **When** a design is first shown publicly — an exhibition, a listing, a lookbook
- **Then** that date is stamped and never editable afterwards
- **Never** backdating it later, which is precisely the field a dispute turns on

R02.6 A rejected sample keeps its reason

- **When** a sample round is rejected
- **Then** the reason is recorded against the version
- **Never** closing it with a status alone, which loses the only information that stops the same mistake in the next round

R02.7 A specification cannot be deleted while stock exists against it

- **When** someone removes a design that has ever been made
- **Then** it is archived and stays linked to every piece produced from it
- **Never** orphaning finished stock from the specification it was made to

Module 03 · Inventory & Catalog — 14 rules

R03.1 Stock is one number per SKU, per location, per stage

- **When** any module asks how much there is

- **Then** it reads the one quantity, with the channel recorded on the movement rather than on the stock
- **Never** keeping a separate stock figure per channel — the last piece sold on one marketplace has to vanish from the other ten at the same instant, which per-channel inventory cannot do

R03.2 Negative stock is a fault, not a state

- **When** an issue would take a quantity below zero
- **Then** the issue is refused
- **Never** recording a negative balance and leaving someone to explain it at month-end

R03.3 Selling a kit decrements every component

- **When** a kit or combo SKU is sold
- **Then** each component SKU is decremented at order time
- **Never** decrementing only the kit, which leaves the components sellable twice

R03.4 A kit with no components is refused

- **When** an item is marked a kit but lists nothing
- **Then** the record is refused and named
- **Never** accepting it and silently decrementing nothing on every sale

R03.5 Stock value ties to the item cost, always

- **When** stock is valued
- **Then** the value is computed from the quantity and the item cost
- **Never** storing a valuation that can drift from the quantity it is supposed to describe

R03.6 Every movement has a source, a destination, or both

- **When** a stock movement is recorded
- **Then** at least one end is named
- **Never** accepting a movement from nowhere to nowhere, which is how quantity appears without a cause

R03.7 A quantity is a whole number above zero

- **When** a movement is written
- **Then** a non-integer or non-positive quantity is refused
- **Never** accepting a negative movement as a shorthand for a reversal — a reversal is its own movement with its own reason

R03.8 Goods in someone else's warehouse are still yours

- **When** stock sits in a channel's own warehouse under consignment or sale-or-return
- **Then** that warehouse is a location like any other and the stock is counted, valued and aged there
- **Never** letting it drop off the books until it sells, which understates both stock and exposure

R03.9 Fabric in metres and pieces in numbers share one item master

- **When** an item is defined
- **Then** its unit of measure is a property of the item
- **Never** building a second item master for a second unit, which splits the one stock number this module exists to protect

R03.10 A listing needs the packed size and weight before it can go out

- **When** a product is pushed to a channel
- **Then** packed dimensions and weight must be present
- **Never** listing without them, because that is the field every courier weight dispute is settled on

R03.11 The channel's own code for a product is mapped, not assumed

- **When** a product exists on a marketplace
- **Then** that channel's identifier is stored against ours
- **Never** matching on name or on a code we invented, which mis-posts every settlement line for that product

R03.12 A duplicate master record is merged, never left as two

- **When** the same customer, vendor or design is detected twice
- **Then** they are merged and both old identifiers keep resolving
- **Never** leaving two live records, which splits every total that record appears in

R03.13 A price is per channel and dated

- **When** a channel price is set
- **Then** it is stored against that channel with the date it takes effect
- **Never** holding one price and reading it as if it applied everywhere and always

R03.14 Dead stock is named as dead stock

- **When** an item has not moved for the period set for it
- **Then** it appears on the dead-stock register with its age and carrying value
- **Never** leaving it inside the general stock figure where it reads as healthy inventory

Module 04 · CRM — 9 rules

R04.1 One customer, one record, whichever channel they arrived by

- **When** the same person orders on a marketplace and later at the counter
- **Then** both land on one record with the channel noted on each order
- **Never** creating a second customer per channel, which makes lifetime value meaningless

R04.2 A document is filed against the record it belongs to

- **When** any agreement, receipt, certificate or scan is stored
- **Then** it is attached to the order, party, case or employee it concerns
- **Never** filing it in a folder that has to be remembered rather than found

R04.3 A signed copy files itself back

- **When** a document sent for signature is signed
- **Then** the signed version returns to the same record automatically
- **Never** leaving the signed copy in an inbox while the record still shows it as pending

R04.4 A ticket carries the order it is about

- **When** a question arrives by chat, email or phone
- **Then** it is tied to the order or account it concerns, with the history already on screen
- **Never** opening a ticket with no link, which makes the first reply a request to explain again

R04.5 Feedback attaches to the item, not only the buyer

- **When** a rating or complaint arrives after delivery
- **Then** it is attached to the design or item it is actually about
- **Never** holding it only against the customer, which hides a complaint-prone item as a scatter of unrelated gripes

R04.6 A customer's consent travels with their data

- **When** a customer record is used for marketing or profiling
- **Then** the consent captured at the point it was given is checked first
- **Never** assuming that having the data implies permission to use it for anything

R04.7 A merged customer keeps both histories

- **When** two customer records are merged
- **Then** every order, ticket and document from both survives on the surviving record
- **Never** discarding the shorter history to make the merge simple

R04.8 Credit state is read at the moment of the order

- **When** a B2B order is placed
- **Then** the customer's outstanding and limit are evaluated then
- **Never** using a figure cached from the last sync, which is how a party goes past its limit between refreshes

R04.9 A closed ticket keeps what resolved it

- **When** a ticket is closed
- **Then** the resolution is recorded on it
- **Never** closing with a status alone, which loses the answer the next identical question needs

Module 05 · Sales — 18 rules

R05.1 Every sale carries its company and its channel

- **When** an order is created on any channel
- **Then** both are written on the order
- **Never** leaving either to be inferred later from the document number or the warehouse

R05.2 A sale posts stock and ledger together

- **When** a sale is confirmed
- **Then** stock is deducted, the invoice is raised, and the ledger is posted in one transaction
- **Never** invoicing without moving stock, or moving stock without posting

R05.3 If the ledger refuses, the stock never moved

- **When** the posting half of a sale fails
- **Then** the stock movement is rolled back with it
- **Never** leaving the goods gone and the books untouched

R05.4 A quote becomes an order without being retyped

- **When** a quotation is accepted
- **Then** the order is created from it, carrying the same lines and prices
- **Never** re-entering the lines, which is where the price on the quote and the price on the invoice start to differ

R05.5 A price below the floor needs an approval, not a note

- **When** a line is priced under the floor set for it
- **Then** the order waits in the approvals queue with the rule that stopped it named
- **Never** letting it through with a comment box, which is a discount policy nobody can enforce

R05.6 An export invoice knows it is an export

- **When** an order ships outside the country
- **Then** the LUT or IGST treatment, currency and shipping terms are set on the order itself
- **Never** treating it as a domestic invoice and correcting the tax afterwards

R05.7 A counter sale is the same order record

- **When** someone buys at the counter
- **Then** the same order table records it, with the counter as the channel
- **Never** running the till on a separate book that has to be merged later

R05.8 A credit sale reserves the credit at the moment it is taken

- **When** a B2B order is accepted on credit
- **Then** the exposure is committed against the party immediately
- **Never** counting it only when the invoice is raised, which lets several orders each fit inside the same limit

R05.9 A dispatch cannot exceed what was ordered

- **When** a shipment is prepared
- **Then** quantities are checked against the order line
- **Never** shipping over, which becomes an invoice the customer never agreed to

R05.10 A cancelled order releases what it held

- **When** an order is cancelled
- **Then** reserved stock and committed credit are both released
- **Never** leaving stock reserved against a dead order, which shows the business as out of goods it actually has

R05.11 An AWB belongs to the shipment, not the courier integration

- **When** a tracking number is recorded, typed in or fetched
- **Then** it is stored on the shipment
- **Never** making the number reachable only through whichever courier API produced it, which loses it the day that courier is dropped

R05.12 A subscription renewal is a new order

- **When** a subscription renews
- **Then** a fresh order is created with its own stock, invoice and posting
- **Never** extending the original order, which makes the revenue of two periods indistinguishable

R05.13 A sale to a sister company is marked as one

- **When** the counterparty is another company in the group
- **Then** the counterparty company is recorded on the entry
- **Never** posting it as an ordinary outside sale, which inflates the group turnover by trade it never did

R05.14 A quote or proforma number carries its type and financial year

- **When** a quotation or proforma is raised
- **Then** it is numbered Q-{FY}-#### or PI-{FY}-####, sequential within that company and year
- **Never** sharing one sequence between quotations and proformas, which makes a proforma indistinguishable from a quote in the register

R05.15 A quote line with no description, no quantity or a negative rate is not a line

- **When** a quotation is totalled
- **Then** only lines with a description, a quantity above zero and a rate of zero or more are counted
- **Never** letting a half-filled row contribute a number to the total

R05.16 An export line carries no GST

- **When** a quotation or invoice is marked export under LUT
- **Then** the GST percentage is zero and the document says why
- **Never** applying the domestic rate and correcting it after the buyer queries the total

R05.17 A made-to-measure order has two money legs, and both are visible

- **When** a customisation order is accepted
- **Then** the advance and the balance are recorded as separate amounts with their own dates, and the balance stays owed until dispatch
- **Never** showing one payment at the end, which hides money already taken and work already owed

R05.18 A customisation quote keeps every round of the negotiation

- **When** a price is revised during a bespoke enquiry
- **Then** each quoted figure is kept in order with what changed
- **Never** overwriting the earlier figure, which is the one the customer remembers agreeing to

Module 06 · Planning & Requirements (MRP) — 8 rules

R06.1 A forecast is labelled a forecast wherever it appears

- **When** a projected figure is shown beside actuals
- **Then** it is visually and structurally distinct
- **Never** letting a forecast total sit in the same column as a real one, which is how a plan becomes a reported result

R06.2 A requirement run reads live stock, not a snapshot

- **When** the MRP run explodes requirements
- **Then** it reads the current quantity at the moment it runs
- **Never** planning against a nightly copy, which orders material the business already has

R06.3 A requirement names what caused it

- **When** the run produces a shortfall
- **Then** the order, forecast or reorder level that generated it is recorded on the line
- **Never** producing a bare quantity nobody can trace back to a demand

R06.4 Stock already on order counts against the shortfall

- **When** the run computes what to buy
- **Then** open purchase orders are netted off first
- **Never** ignoring them and ordering the same material twice

R06.5 A budget ceiling refuses, it does not warn

- **When** a proposed purchase would exceed the open-to-buy ceiling
- **Then** it is held for approval with the ceiling named
- **Never** raising it with a warning, which makes the ceiling advisory and therefore not a ceiling

R06.6 A lead time is per vendor and per item

- **When** a run works out when to order
- **Then** it uses the lead time recorded for that vendor and that item
- **Never** applying one global lead time, which under-orders the slow lines and over-orders the fast ones

R06.7 A run is kept, not overwritten

- **When** the MRP run executes again
- **Then** the previous run stays readable with its inputs

- **Never** replacing it, which makes it impossible to see why last week's decision was taken

R06.8 A seasonal signal cannot silently become a permanent one

- **When** a festival or season inflates demand
- **Then** the period it applies to is stored with the signal
- **Never** folding a spike into the baseline, which keeps ordering for a festival all year

Module 07 · Purchase — 12 rules

R07.1 Nothing is paid without a three-way match

- **When** a vendor invoice is approved
- **Then** the purchase order, the goods received note and the invoice must agree
- **Never** paying on the invoice alone, which pays for goods that never arrived

R07.2 A short or damaged receipt is recorded as received short

- **When** the GRN quantity is below the PO quantity
- **Then** the difference is recorded with its reason and the payable follows the received quantity
- **Never** receiving the full quantity to make the match pass

R07.3 Input tax credit is claimed against a real document

- **When** ITC is taken on a purchase
- **Then** the vendor invoice and its tax detail are on file
- **Never** claiming credit from a payment record alone, which is the claim that fails reconciliation

R07.4 Landed cost reaches the item, not just the P&L

- **When** freight, duty or insurance is attached to a purchase
- **Then** it is apportioned into the cost of the items received
- **Never** expensing it separately, which understates the cost of every piece made from that material

R07.5 A vendor price is dated

- **When** a rate is agreed with a supplier
- **Then** it is stored with the date it takes effect
- **Never** overwriting the old rate, which makes last month's purchase look mispriced

R07.6 A purchase order over its approval level waits

- **When** a PO exceeds the value a role may approve
- **Then** it goes to the approvals queue naming the rule and the level
- **Never** splitting it into smaller orders to fit under the limit — the split is detected and the parts are assessed together

R07.7 A vendor with no active record cannot be paid

- **When** a payment is raised

- **Then** the vendor must exist, be active, and have its bank detail verified
- **Never** paying to detail typed onto the payment itself, which is the single most common route for payment fraud

R07.8 A change to vendor bank detail is treated as high risk

- **When** a vendor's bank account is changed
- **Then** the change is approved by a second person and the old detail is kept
- **Never** accepting a change from an email instruction alone

R07.9 A job-work despatch stays on the books

- **When** material is sent to a contractor
- **Then** it moves to a job-work location and remains this company's stock
- **Never** writing it out on despatch, which loses material the business still owns

R07.10 An insurance policy is linked to what it covers

- **When** a policy is recorded
- **Then** the stock, premises or shipment it covers is named on it
- **Never** holding policies as documents with no link, which is discovered only at the moment of a claim

R07.11 The three-way match is arithmetic, not a judgement

- **When** a vendor invoice is checked
- **Then** the payable equals the received quantity $\times$ the purchase-order rate, and the purchase order, the goods receipt and the invoice must all agree on quantity and value
- **Never** passing an invoice whose value exceeds received quantity $\times$ agreed rate, and never letting an override happen without recording who made it and why

R07.12 A material is sourced down a ranked list, not from whoever answers

- **When** a material has to be bought
- **Then** the vendors ranked for that material are approached in their priority order
- **Never** defaulting to the last vendor used, which is how a price rise becomes permanent without anyone deciding

Module 08 · Manufacturing — 20 rules

R08.1 Sets are pooled across every maker before the minimum is taken

- **When** completed sets are counted for a design
- **Then** every maker's pieces for that design are pooled first, and the set count is the minimum across the populated member columns of the pool
- **Never** counting sets per maker row and adding them up, which loses every set completed by two people between them

R08.2 A surplus piece is paid for, and is not a set

- **When** a maker produces more of one component than the set needs

- **Then** the extra is named individually and paid at its own piece rate
- **Never** adding it to the set count, and never leaving it unpaid because it did not complete a set — the person made it either way

R08.3 A design counts on the components it actually has

- **When** a design is made of fewer component types than the usual set
- **Then** it is counted on the members it does have
- **Never** returning zero because an optional member is absent, which silently unpays a whole design

R08.4 A missing rate posts zero and is flagged, never guessed

- **When** a design has no entry in the piece-rate master
- **Then** it costs zero and the design is named in the summary
- **Never** inferring a rate from a similar design — a guessed rate is a wrong payment to a real person

R08.5 A two-row heading is read as two rows

- **When** the production grid uses a merged heading over component columns
- **Then** both header rows are read so repeated component names stay distinct columns
- **Never** reading only the first row, which collapses three same-named component columns into one and undercounts the work

R08.6 A worker written as a pair stays one unit

- **When** two names share one row as a working pair
- **Then** they are treated as a single paying unit
- **Never** splitting them into two workers, which halves each person's recorded output and breaks the payout

R08.7 Several years of grids pool into one set of figures

- **When** more than one production workbook is supplied
- **Then** their grids pool into a single costing
- **Never** reporting each file separately, which double-counts nothing but hides the sets completed across a year boundary

R08.8 Cost per piece is independent of set completion

- **When** the cost of a design is worked out
- **Then** each raw piece is costed at its own rate
- **Never** costing by completed sets, which values an unfinished set at nothing while the labour has already been spent

R08.9 A production report moves stock and pay together

- **When** a piece-work production report is accepted
- **Then** finished stock comes in, the payout is raised in HR, wages post to the ledger, and the design cost updates — in one transaction

- **Never** taking the stock in and settling the pay in a separate pass, which is how the two disagree

R08.10 Material issued to production leaves raw stock at the moment it is issued

- **When** a production order consumes material
- **Then** raw stock is reduced and work in progress increases
- **Never** consuming at completion, which shows material as available while it is already cut

R08.11 A bill of materials is versioned with the design

- **When** a production order is created
- **Then** it captures the BOM version in force at that moment
- **Never** reading the current BOM when costing an old order, which recosts history

R08.12 Wastage is recorded, not absorbed

- **When** consumption exceeds the BOM
- **Then** the excess is recorded as wastage against the order with its reason
- **Never** quietly increasing the BOM to match what was used, which destroys the only signal that something is going wrong

R08.13 A stage cannot be skipped without being recorded as skipped

- **When** work moves past a defined stage without that stage being marked
- **Then** the skip is recorded on the order
- **Never** letting the stage silently complete, which makes every stage-time figure fiction

R08.14 An advance to a worker is a balance, not a deduction from nowhere

- **When** an advance is paid
- **Then** it is held against that worker and recovered from later payouts, with the running balance visible
- **Never** deducting an amount at payout time that cannot be traced to a specific advance

R08.15 A rework carries the cost of the rework

- **When** a piece is returned to a stage to be redone
- **Then** the additional labour is costed to the design that caused it
- **Never** costing it as new production, which makes a failing design look as profitable as a good one

R08.16 Material consumed is the average per piece times the pieces made

- **When** consumption is costed against a production run
- **Then** consumption equals the average consumption per piece $\times$ pieces produced, and the difference against the bill of materials is recorded as wastage
- **Never** back-fitting the average to whatever was issued, which makes wastage mathematically impossible to see

R08.17 A set type comes from the rate master, and an inferred one says so

- **When** a design is classified into a set type

- **Then** the rate master's Set column decides it; when the design is absent, the type is inferred from which component columns actually carry pieces and the design is flagged as inferred
- **Never** presenting an inferred classification as though it came from the master

R08.18 An alteration caused by the worker's own mistake is unpaid

- **When** a piece is reworked because of an error by the person who made it
- **Then** the alteration hours are recorded and paid at zero
- **Never** paying for the rework at the standard alteration rate, and never leaving the hours unrecorded — the time still happened and the design still bore the cost

R08.19 Alteration time is paid at the alteration rate, not the piece rate

- **When** admin-assigned alteration hours are settled
- **Then** they are paid at the hourly alteration rate in force and added to that worker's payout
- **Never** folding alteration hours into the piece count, which corrupts both the production figure and the earnings figure at once

R08.20 A contract worker paid by the hour has no attendance row

- **When** a contract role is settled
- **Then** payment is hours worked × the agreed hourly rate, recorded against the person without an attendance record
- **Never** forcing a contract worker through the salaried attendance model, which produces a monthly figure nobody agreed to

Module 09 · Quality & Compliance — 7 rules

R09.1 A failed check blocks the next stage

- **When** an inspection fails
- **Then** the batch cannot progress until it is passed, reworked or written off
- **Never** letting it move with the failure noted, which sends a known defect to a customer

R09.2 A check names the person who did it

- **When** any inspection is recorded
- **Then** the inspector, the time and the sample size are stored
- **Never** accepting an anonymous pass, which cannot be investigated when the complaints arrive

R09.3 An expiring certificate warns before it expires

- **When** a certificate approaches its expiry
- **Then** it is raised while there is still time to renew
- **Never** discovering the lapse at the moment a buyer asks for it

R09.4 A rejected batch cannot be sold as first quality

- **When** a batch is rejected

- **Then** it is marked and can only be sold through a channel that accepts seconds
- **Never** letting it re-enter the ordinary sellable pool

R09.5 A defect is attached to the design and the stage

- **When** a defect is recorded
- **Then** both the design and the stage that produced it are named
- **Never** recording it against the batch alone, which loses the pattern that would have prevented the next one

R09.6 A compliance document is evidence, not a checkbox

- **When** a compliance requirement is marked met
- **Then** the document proving it is attached
- **Never** accepting a tick with nothing behind it, which is what fails an audit

R09.7 A sustainability figure comes from the same evidence

- **When** an ESG figure is reported
- **Then** it is computed from the certificate and audit records already on file
- **Never** assembling it separately once a year from numbers nobody can trace

Module 10 · Warehouse — 8 rules

R10.1 A pick is confirmed against the bin it came from

- **When** an item is picked
- **Then** the bin is recorded on the movement
- **Never** decrementing a warehouse total with no bin, which makes the next cycle count unexplainable

R10.2 A short pick stops the pack, it does not silently reduce the order

- **When** the picker cannot find the full quantity
- **Then** the shortage is raised against the order and the pack waits
- **Never** packing what was found and invoicing for it as though that was the order

R10.3 A scan is the same event as a keyed entry

- **When** a code is captured by scanner, phone camera or typing
- **Then** the same movement is written
- **Never** having a scanning path that writes different records from the manual path

R10.4 A cycle count adjustment names a reason

- **When** a count differs from the system
- **Then** the adjustment records the reason and the person
- **Never** writing the system down to the counted figure with no explanation, which hides theft and damage equally well

R10.5 The packing video is linked to the shipment

- **When** a parcel is recorded on video at packing
- **Then** the recording is attached to that shipment
- **Never** keeping the footage in a folder by date, which makes it unusable in the dispute it exists for

R10.6 A bin holds a location, not a guess

- **When** stock is put away
- **Then** the destination bin is captured at put-away
- **Never** assigning a default bin so the step can be skipped

R10.7 A dispatch cut-off is per channel

- **When** a channel has a handover deadline
- **Then** the queue is ordered and warned against that channel's own cut-off
- **Never** applying one cut-off to all of them, which misses the earliest and idles for the latest

R10.8 A returned parcel is inspected before it is anything else

- **When** a return arrives at the warehouse
- **Then** it is booked into a return-inspection location first
- **Never** restocking on arrival, which puts an unchecked item back on sale

Module 11 · Logistics — 11 rules

R11.1 The courier rate is checked against the packed weight

- **When** a courier bills for a shipment
- **Then** the billed weight is compared with the packed weight recorded at packing
- **Never** accepting the courier's weight without comparison, which is the most consistently overcharged line in the business

R11.2 A weight dispute is raised with the evidence attached

- **When** billed and packed weight differ beyond tolerance
- **Then** a dispute is raised carrying the packing record
- **Never** absorbing the difference because each one is small

R11.3 An undelivered parcel is chased before it becomes a return

- **When** a delivery attempt fails
- **Then** the NDR is actioned within the window the courier allows
- **Never** letting it lapse into a return, which costs the freight twice and the sale once

R11.4 COD collected is a receivable until it is remitted

- **When** a COD parcel is delivered
- **Then** the amount is a receivable from the courier
- **Never** treating delivery as payment, which reports cash the business does not have

R11.5 A remittance is matched parcel by parcel

- **When** a courier remits COD
- **Then** each parcel in the remittance is matched individually
- **Never** accepting the total, which is how short remittances go unnoticed for months

R11.6 A manifest is a record, not a printout

- **When** parcels are handed over
- **Then** the handover is recorded against each shipment with the time and the person
- **Never** keeping only a signed sheet, which cannot be queried when a parcel is disputed

R11.7 An RTO parcel is stock again only after inspection

- **When** a return to origin is received
- **Then** it goes through inspection before it can be sold
- **Never** restocking it automatically on scan

R11.8 Freight cost reaches the order it belongs to

- **When** a shipment is costed
- **Then** the freight is attributed to the order
- **Never** holding freight only as a monthly expense, which makes per-order and per-channel profit fiction

R11.9 A courier can be changed without losing history

- **When** a courier is switched off
- **Then** every past shipment, AWB and dispute stays readable
- **Never** making history depend on an integration that is still connected

R11.10 A zone and rate card are dated

- **When** courier rates change
- **Then** the new card is stored with its effective date
- **Never** overwriting the card, which makes every past shipment look mischarged

R11.11 A partial-COD order has two collections and both are tracked

- **When** an order is placed with an advance online and the balance on delivery
- **Then** the advance is a receipt now and the balance is a receivable from the courier until it is remitted
- **Never** treating the advance as the whole payment, which makes every such order look settled while most of the money is still outstanding

Module 12 · Accounting & GST — 24 rules

R12.1 Money is an integer count of paise

- **When** any amount is held, added or compared
- **Then** it is an integer number of paise, becoming a decimal string only where a person reads it

- **Never** holding money in a floating-point number, where ₹0.10 + ₹0.20 is not ₹0.30 and a trial balance stops balancing

R12.2 An amount finer than a paisa is refused, not rounded

- **When** a computation produces a fraction of a paisa
- **Then** it is refused and the caller must round deliberately
- **Never** rounding silently, which is how two sides of the same figure drift apart and nobody can say which is right

R12.3 A split sums back to the original, exactly

- **When** an amount is divided — across lines, across companies, across periods
- **Then** the parts add back to the whole, with the round-off returned as its own figure
- **Never** losing or inventing a paisa in the split, and never hiding the remainder inside the largest part

R12.4 An unbalanced entry is refused, with the gap named

- **When** a voucher is posted whose debits and credits differ
- **Then** it is refused and the difference is stated
- **Never** posting it to a suspense account to make it balance, which converts an error into a permanent record

R12.5 A line cannot be a debit and a credit at once

- **When** a posting line carries both
- **Then** it is refused
- **Never** netting the two into whichever is larger

R12.6 The trial balance is computed, never stored

- **When** the trial balance is asked for
- **Then** it is summed from the posting lines at that moment
- **Never** reading a maintained total, which is a number that can be wrong without anything looking wrong

R12.7 A locked period refuses a backdated entry

- **When** a voucher is dated inside a closed period
- **Then** it is refused and the lock that stopped it is named
- **Never** posting it into the current period instead, which silently moves last year's result into this one

R12.8 Unlocking a period is itself recorded

- **When** a closed period is reopened
- **Then** who reopened it, when and why is written to the trail
- **Never** allowing a quiet reopen, which is the one action that could undo every other guarantee here

R12.9 A tax rate resolves on the date of the document

- **When** tax is computed for any invoice

- **Then** the rate in force on that document's date is used
- **Never** applying today's rate to an old invoice, which makes correct history look like an error

R12.10 Two rates covering one date is ambiguous, not a coin toss

- **When** two effective-dated rows overlap for the same date
- **Then** the resolution is refused and the overlap is named
- **Never** picking the newer one, which makes the answer depend on insertion order

R12.11 A voided entry is reversed, never erased

- **When** a posted voucher is wrong
- **Then** a reversing entry is posted and both stay visible
- **Never** editing or deleting the original, which is the difference between a correction and a cover-up

R12.12 Every figure clicks down to the record that produced it

- **When** any total appears on any screen
- **Then** it is a live query that can be opened down to its vouchers and their documents
- **Never** showing a figure that cannot be traced — an untraceable number is a defect, not a rounding difference

R12.13 An invoice number is sequential per company and per series

- **When** an invoice is raised
- **Then** it takes the next number in that company's series
- **Never** reusing, skipping or back-filling a number, which is the first thing a tax audit tests

R12.14 A GST return is built from vouchers, not from a summary

- **When** GSTR-1 or 3B is prepared
- **Then** it is computed from the underlying invoices
- **Never** accepting a typed summary figure, which cannot be reconciled when the portal disagrees

R12.15 ITC is claimed only where the supplier has filed

- **When** input credit is taken
- **Then** it is matched against the supplier's filed data and the unmatched part is held
- **Never** claiming everything and reversing later, which turns a reconciliation into a liability

R12.16 A place of supply decides the tax, not the billing address

- **When** GST is computed
- **Then** the place of supply determines CGST/SGST or IGST
- **Never** defaulting to the billing address, which mis-splits the tax on every drop-ship

R12.17 A credit note references the invoice it reverses

- **When** a credit note is raised
- **Then** the original invoice is named on it

- **Never** issuing a free-standing credit note, which cannot be matched in either set of books

R12.18 Depreciation is posted, not just calculated

- **When** a period closes
- **Then** depreciation is posted as an entry like any other
- **Never** showing it as a computed figure on a report while the ledger disagrees

R12.19 A company with no tax registration is still a company

- **When** a group company has no registration of its own
- **Then** it keeps its own books and joins the group figures
- **Never** dragging it into a return it does not belong in, and never leaving it out of the group result

R12.20 Year-end close locks, and the lock is the record

- **When** a financial year is closed
- **Then** the period is locked and the closing balances are carried forward as an entry
- **Never** leaving the year open indefinitely so late entries can drift in unnoticed

R12.21 Every voucher type posts through one engine

- **When** a sale, purchase, credit note, debit note, payment, receipt, journal, contra or counter sale is recorded
- **Then** all nine post through the same ledger routine
- **Never** giving a voucher type its own posting logic — this is where home-built accounting breaks and the modules stop agreeing about the same figure

R12.22 Net GST is input against output, per period, per company

- **When** the GST position for a period is computed
- **Then** it is output tax less eligible input credit for that company and that period
- **Never** netting across companies, which offsets one registration's liability with another's credit and is not a return anyone may file

R12.23 Money never becomes a float, in any layer

- **When** an amount is stored, moved between the engine and the database, or exported
- **Then** it stays an integer count of paise end to end, converted for display only
- **Never** a real, double, float or an unlabelled decimal column anywhere a money value lives

R12.24 A money column says what unit it is in

- **When** a column holds an amount
- **Then** its name ends in paise
- **Never** a column called total, amount or cost with no unit — the same name read as rupees by one developer and paise by the next is a factor of a hundred in the books

Module 13 · Treasury & Financial Planning — 8 rules

R13.1 A forecast never posts to the ledger

- **When** a cash-flow projection is produced
- **Then** it is held as a projection, separate from posted entries
- **Never** writing an expected receipt into the books, which reports money that has not arrived

R13.2 A bank line is matched to a voucher, not to a total

- **When** a bank statement is reconciled
- **Then** each line is matched to the entry that caused it
- **Never** reconciling on the closing balance alone, which hides two errors that happen to cancel

R13.3 An unmatched bank line stays visible until it is explained

- **When** a statement line cannot be matched
- **Then** it stays on the unreconciled list with its age
- **Never** writing it off to a sundry account to clear the screen

R13.4 A PDC is a commitment before it is cash

- **When** a post-dated cheque is received
- **Then** it is tracked as a commitment until it clears
- **Never** recognising it as cash on receipt

R13.5 Budget versus actual compares like with like

- **When** a variance is shown
- **Then** both sides use the same period, company and account basis
- **Never** comparing a full-year budget against a part-year actual without saying so

R13.6 A cash forecast names its assumptions

- **When** a projection is produced
- **Then** the collection and payment assumptions behind it are stored with it
- **Never** presenting a projection whose basis cannot be recovered a month later

R13.7 Inter-company funding is recorded on both sides

- **When** one group company funds another
- **Then** both companies post it, naming each other as counterparty
- **Never** recording it in one set of books only, which leaves the group permanently out of balance

R13.8 A currency amount keeps the rate it was converted at

- **When** a foreign-currency transaction is recorded
- **Then** the original amount, the currency and the rate used are all stored
- **Never** storing only the converted figure, which cannot be revalued or explained afterwards

Module 14 · Settlement — 13 rules

R14.1 A payout is matched line by line to orders

- **When** a marketplace settlement file arrives
- **Then** every line is matched to the order it belongs to
- **Never** accepting the net credited amount, which is how a short payment becomes invisible

R14.2 Every deduction is identified before the payout is accepted

- **When** commission, shipping, penalty, TCS or TDS is deducted
- **Then** each is posted to its own account
- **Never** posting the deductions as one lump, which makes an overcharge impossible to find

R14.3 A variance beyond tolerance raises a claim

- **When** the settled amount differs from the expected amount
- **Then** a claim is raised carrying the order, the expectation and the difference
- **Never** absorbing it because it is small — the small ones are the recurring ones

R14.4 A claim has a deadline and the deadline is tracked

- **When** a claim is raised
- **Then** the channel's filing window is stored and warned on
- **Never** letting a valid claim expire unfiled

R14.5 An expected settlement exists from the moment of the sale

- **When** an order is confirmed on a marketplace
- **Then** a settlement expectation is created then
- **Never** waiting for the payout to discover what should have arrived

R14.6 TCS and TDS are receivables, not costs

- **When** a marketplace deducts tax at source
- **Then** it is posted as a receivable against the tax authority
- **Never** expensing it, which understates profit and loses the credit

R14.7 A settlement is reconciled to the bank, not just to the file

- **When** a payout is recorded
- **Then** it is matched to the actual bank credit
- **Never** treating the settlement report as proof that the money arrived

R14.8 A re-sent settlement file does not double-post

- **When** the same settlement file is imported twice
- **Then** already-matched lines are recognised and skipped
- **Never** posting them again, which doubles both revenue and deductions

R14.9 A fee schedule is dated and compared against

- **When** a commission is deducted
- **Then** it is checked against the agreed rate in force on that date
- **Never** accepting whatever rate the file states, which is the single largest silent leak in marketplace trade

R14.10 A settled order is profitable or unprofitable at the SKU

- **When** a payout is fully matched
- **Then** the true net per SKU is computed after every deduction
- **Never** judging profitability on the listed price, which ignores the third of it that never arrives

R14.11 A claim that is paid closes against the original variance

- **When** a channel credits a claim
- **Then** it is matched back to the variance it settles
- **Never** posting the credit as unrelated income, which leaves the variance open forever

R14.12 A settlement figure never overwrites a sale figure

- **When** the settlement disagrees with the order
- **Then** both are kept and the difference is the variance
- **Never** adjusting the original sale to match the payout, which erases the evidence of the shortfall

R14.13 The realisation on a marketplace sale is the price minus every deduction

- **When** what a channel sale actually earned is computed
- **Then** it is the selling price less shipping, commission, fixed fee, GST on those fees, TCS and TDS — each taken from the settlement file
- **Never** judging a sale on its listed price, which ignores the part of it that never arrives, and never applying an assumed commission percentage when the file states the real one

Module 15 · E-commerce / OMS — 19 rules

R15.1 Companies and channels are read from the data, never from a list in the code

- **When** orders or sheets from any number of companies and channels are processed
- **Then** the companies and channels present are discovered and each gets its own columns
- **Never** writing a fixed set of companies or channels into the software, which caps the business at whatever it happened to have on the day the code was written

R15.2 A tenth or eleventh channel needs no code change

- **When** a new marketplace or company is added
- **Then** it is a row, and every figure, column and consolidation follows
- **Never** requiring a release to sell somewhere new

R15.3 A channel belongs to a company

- **When** two companies both sell on the same marketplace
- **Then** each has its own channel record, and both may use the same short code
- **Never** sharing one channel across companies, which merges two companies' sales into one figure

R15.4 A price is never invented for an item that has none

- **When** an item has no price on file
- **Then** it is reported as having no price and named in the summary
- **Never** substituting an average or a similar item's price, which quietly fabricates revenue

R15.5 Net is sale minus return, and inventory is net plus wrong return

- **When** quantities are rolled up
- **Then** net sale is sale minus return, and the inventory figure adds back the wrong returns
- **Never** treating a wrong return as ordinary saleable stock, because it is not the item that was sent out

R15.6 A blank cell is blank, not a value

- **When** a column contains only whitespace
- **Then** it is read as empty
- **Never** treating a lone space as a marked entry, which converts formatting into business fact

R15.7 An item that only ever came back is still reported

- **When** an item has returns but no sales in the period
- **Then** it appears with its returns
- **Never** dropping it because it has no sale line, which hides the worst-performing items entirely

R15.8 A totals row is the sum of the rows above it

- **When** a report shows a total
- **Then** it equals the rows it sits under
- **Never** computing the total by a different route from the detail, which is how a report disagrees with itself

R15.9 A marketplace order pull creates a real order

- **When** orders are fetched from a channel
- **Then** a sales order is created, stock is reserved, and the pick list follows
- **Never** holding channel orders in a staging area that has to be re-entered to become real

R15.10 A cancelled channel order releases its reservation

- **When** the channel cancels an order
- **Then** the reservation is released and the cancellation recorded
- **Never** leaving stock reserved against an order the channel has already dropped

R15.11 A wrong return is dead stock, not stock

- **When** a return is inspected and found to be a different or damaged item

- **Then** it is written to dead stock with its cost recognised as a loss
- **Never** restocking it as first quality, which sells a customer the same problem twice

R15.12 A listing rejected by a channel says why

- **When** a push to a channel fails
- **Then** the rejection and its reason are reported back against the listing
- **Never** reporting a push as successful when part of it failed, which leaves the business believing it is present where it is not

R15.13 A manual data check is a recorded step, not a habit

- **When** figures are checked by hand before a cycle closes
- **Then** the check, the person and the outcome are recorded
- **Never** relying on someone remembering to look

R15.14 A channel-specific SKU code never becomes the master code

- **When** a channel uses its own identifier
- **Then** it is stored as a mapping against our SKU
- **Never** adopting the channel's code as the item code, which breaks the moment a second channel does the same

R15.15 A size recommendation is advice, never a silent substitution

- **When** a fit suggestion is offered
- **Then** it is shown as a recommendation the customer chooses
- **Never** changing the size on an order on the customer's behalf

R15.16 An order held past its cut-off is escalated, not queued

- **When** an order approaches the channel's dispatch deadline
- **Then** it is raised to the person who can act, naming the deadline
- **Never** letting it age quietly into a penalty

R15.17 Closing stock is opening plus in minus out

- **When** a stock position is computed for a period
- **Then** closing = opening + receipts – issues, from the movements themselves
- **Never** carrying a maintained closing figure that can drift from the movements that produced it

R15.18 Courier return, customer return and wrong return cost three different things

- **When** a return is processed
- **Then** a courier return costs repacking only, a customer return costs alteration plus iron plus packing at the rate set for that design, and a wrong return is written off at the full selling price
- **Never** applying one blended return cost to all three, which hides the expensive kind inside the cheap kind

R15.19 A wrong return is never added back to stock

- **When** a return is found to be a different item from the one sent
- **Then** it becomes dead stock and the selling price is recognised as a loss
- **Never** restocking it, at any value, however sellable it looks

Module 16 · HR & Payroll — 22 rules

R16.1 A raise closes the old row, it does not overwrite it

- **When** a salary or rate changes
- **Then** the row in force is closed on the day before, and a new row opens
- **Never** editing the existing figure, which rewrites what the person was actually paid last year

R16.2 History resolves to what was actually in force

- **When** a past month is recomputed
- **Then** the rate in force in that month is used
- **Never** recomputing an old payslip at today's rate

R16.3 A future-dated raise activates by itself

- **When** a raise is entered with a future date
- **Then** it takes effect when that month arrives, with nobody remembering to apply it
- **Never** requiring a manual step, which is how an agreed raise is missed

R16.4 A month with nothing in force raises, and never returns zero

- **When** no rate covers the month being computed
- **Then** the computation is refused and the gap is named
- **Never** returning zero, which pays a real person nothing and looks like a valid answer

R16.5 Backdating over an open row is refused

- **When** a change is entered with a date inside a period already settled
- **Then** it is refused
- **Never** silently rewriting history that has already been paid and posted

R16.6 A person can leave and come back

- **When** someone rejoins after a break
- **Then** the spell log holds both periods and the gap between them
- **Never** creating a second employee record, which splits their history and their service

R16.7 Month spans handle February and the year end

- **When** a period is computed across month or year boundaries
- **Then** the real calendar is used
- **Never** assuming thirty-day months, which is wrong twelve times a year and badly wrong in February

R16.8 Staff and piece-rate workers sit in one register

- **When** payroll is prepared
- **Then** monthly staff and piece-rate workers are computed in the same run and paid from the same register
- **Never** running two payrolls that have to be added together by hand

R16.9 An advance is recovered against a named advance

- **When** a deduction is made at payout
- **Then** it names the advance it is recovering and reduces that balance
- **Never** deducting an amount that cannot be traced to a specific advance

R16.10 Attendance drives pay, and both are visible together

- **When** a payout is computed
- **Then** the attendance it was computed from is shown beside it
- **Never** presenting a pay figure whose basis the person being paid cannot see

R16.11 Identity documents are read, never stored in a file that leaves

- **When** Aadhaar, PAN, bank or UPI detail is used for a computation
- **Then** it is used and not serialised into any exported or committed artifact
- **Never** writing personal identifiers into a report, a backup file or a repository

R16.12 A payout that fails to post does not mark as paid

- **When** the bank transfer or the ledger posting fails
- **Then** the payout stays unpaid and the failure is raised
- **Never** marking it paid on submission, which loses a real person's wages in the gap

R16.13 The daily rate is the monthly salary divided by twenty-seven

- **When** a day of attendance is priced
- **Then** the daily rate is that month's salary $\div$ 27, using the salary in force in that month
- **Never** using calendar days, working days, or a rate carried over from a month with a different salary

R16.14 Attendance codes have fixed multipliers and a blank is absent

- **When** earned pay is computed from attendance
- **Then** present, holiday, on-duty and paid leave count 1, a half day counts 0.5, absent and unpaid leave count 0, and an empty cell counts as absent
- **Never** treating a blank as present, or as unknown to be filled in later — a blank that pays is a blank that will be left blank

R16.15 Threshold hours do not move when salary moves

- **When** a raise takes effect
- **Then** the monthly hour threshold for that role stays as it was
- **Never** scaling the threshold with the salary, which silently changes what the person is expected to work in exchange for a raise

R16.16 Productivity cost is that month's salary over the threshold, times hours worked

- **When** the cost of a person's time is charged to work
- **Then** it is $(\text{salary in force that month} \div \text{threshold hours}) \times \text{the hours actually active}$
- **Never** using a single annual figure, which misprices every month on either side of a raise

R16.17 A holiday is paid and produces no hours

- **When** a holiday is marked
- **Then** it pays a full day and contributes zero productive hours
- **Never** counting holiday hours as production, which flatters every efficiency figure that reads them

R16.18 A half day is half the hours, from the same start

- **When** a half day is marked
- **Then** it starts at the normal in-time and its hours are half the full shift for that person's pattern
- **Never** assuming a fixed midday finish for everyone, when the male and female shift lengths differ

R16.19 The festival flag drives leave and nothing else

- **When** a religion is recorded against a person
- **Then** it is used only to match a festival-leave request
- **Never** using it as a filter, a grouping or a report dimension anywhere else in the system

R16.20 A geofence failure flags, it does not refuse

- **When** attendance is marked outside the radius set for the unit, or outside the grace window
- **Then** it is recorded with the flag and raised to the manager
- **Never** refusing the mark — a system that locks someone out of being paid for standing at the wrong gate has failed at its actual job

R16.22 A shared document carries the pay rules, never the pay roster

- **When** a plan, a specification or any document that leaves this building is generated
- **Then** it carries the formulas, thresholds and effective-dating that decide pay, and refers to the roster rather than reproducing it
- **Never** printing an individual's name beside their salary, or their religion at all, into a document that is committed to a repository and travels with every copy — the software needs those fields, a reader of the plan does not

R16.21 An override is allowed and is always recorded

- **When** an administrator corrects attendance, a geofence flag or a payroll figure
- **Then** the change, the person and the reason go to the audit trail
- **Never** an override that leaves no trace, which is indistinguishable from the system having been wrong

Module 17 · Marketing — 10 rules

R17.1 A campaign is measured on revenue, not on opens

- **When** campaign performance is reported
- **Then** it is attributed to actual orders
- **Never** reporting opens and clicks as the result, which measures the message rather than the business

R17.2 A repricing rule shows what it did

- **When** a rule changes a price
- **Then** the change, the rule that made it and the effect on orders are recorded together
- **Never** changing prices with no record, which makes a bad rule impossible to identify or reverse

R17.3 A price floor is a floor

- **When** a repricing rule would go below the floor set for a SKU
- **Then** it stops at the floor
- **Never** undercutting to match a competitor below cost

R17.4 A markdown starts before the stock is dead, not after

- **When** stock reaches the age set for it
- **Then** the markdown schedule begins
- **Never** waiting until it is unsellable, which converts a lower-margin sale into a write-off

R17.5 A campaign cannot message someone who has not consented

- **When** a marketing send is prepared
- **Then** the recipient list is filtered by consent at send time
- **Never** sending to a list captured before the consent was checked

R17.6 A published page reads live catalogue data

- **When** a page shows a price or a stock state
- **Then** it reads the same record the order screen reads
- **Never** pasting a figure into the page, which goes stale the first time the price changes

R17.7 An exhibition is a channel

- **When** leads and sales come from a trade show
- **Then** they land in CRM and the order book against that channel
- **Never** collecting them on paper to be entered later, which is where they are lost

R17.8 A marketing automation cannot move money

- **When** a campaign rule fires
- **Then** it may message, tag, schedule or reprice within its limits
- **Never** issuing a refund, a credit note or a payment — that is not what this engine is allowed to do

R17.9 A scheduled post that fails is reported as failed

- **When** a scheduled publication does not go out

- **Then** it is raised with the reason
- **Never** showing it as published in the calendar while nothing was posted

R17.10 Return on ad spend is measured against real orders

- **When** campaign performance is computed
- **Then** it is revenue from attributed orders ÷ spend actually incurred
- **Never** using a platform's own reported conversions as the revenue figure, which counts orders this system has no record of

Module 18 · AI Content Engine — 11 rules

R18.1 Content is written from the catalogue, not about the category

- **When** a listing or description is generated
- **Then** it is generated from that product's own attributes
- **Never** writing plausible copy about the kind of thing it is, which is how a listing describes features the product does not have

R18.2 Structured fields get keywords; anything a human reads gets feeling

- **When** text is produced for a back-end field versus a caption
- **Then** each is written for its own reader
- **Never** writing both the same way, which is the clearest signal of machine-written content

R18.3 Product nouns are banned from creative surfaces

- **When** a caption or a hook is written
- **Then** the product noun is excluded
- **Never** letting search vocabulary bleed into copy meant to be felt

R18.4 The engine criticises its own draft before anyone sees it

- **When** a draft is produced
- **Then** it is put through the self-critique pass and the second draft is what is shown
- **Never** showing the first attempt, which is rarely the best one

R18.5 A render is sought, never recorded

- **When** a video is produced from a page
- **Then** the clock is driven by hand and each frame is captured at its exact instant
- **Never** playing the animation and recording the screen, which bakes whatever else the machine was doing into the customer's reel

R18.6 The same scene renders to the same file

- **When** a render is repeated
- **Then** the output is identical to the byte

- **Never** producing a different file each time, which makes the output impossible to check or approve

R18.7 A generated asset is labelled as generated

- **When** an image or video is produced by a model
- **Then** it carries that fact in the asset record
- **Never** letting a generated image become indistinguishable from a photograph of the actual product

R18.8 Generation stays badged a mockup until a real provider is wired

- **When** a capability is demonstrated without a live provider behind it
- **Then** it is labelled a mockup wherever it appears
- **Never** showing a simulated render as a finished one

R18.9 Image generation states that it needs a graphics card

- **When** the image generation slot is opened with no provider attached
- **Then** it says so plainly and produces nothing
- **Never** presenting a finished-looking screen that cannot generate anything

R18.10 A cloned voice needs the consent of the person it came from

- **When** a voice is cloned for narration
- **Then** that person's recorded consent is on file against them
- **Never** cloning from a recording merely because it was available

R18.11 A publish reports what actually went live

- **When** content is pushed to several destinations
- **Then** each result comes back individually, with reasons for rejections
- **Never** reporting one overall success, which leaves the business absent where it believes it is present

Module 19 · SEO, AEO & AIO — 6 rules

R19.1 Structured data describes what is actually on the page

- **When** schema markup is generated
- **Then** it is generated from the same record the page renders
- **Never** marking up a price or availability that differs from the page, which is penalised and deserved

R19.2 A ranking figure names where it was measured

- **When** a position or citation is reported
- **Then** the engine, the query and the date are stored with it
- **Never** reporting a bare position, which cannot be compared with anything

R19.3 An answer-shaped page still says the same thing as the product record

- **When** content is shaped to be quoted by an answer box

- **Then** the claims match the catalogue
- **Never** writing a more quotable claim than the product supports

R19.4 A technical fix is verified on the live page

- **When** a technical SEO issue is marked resolved
- **Then** the live page is re-fetched and re-checked
- **Never** closing it because the change was deployed

R19.5 AI-engine visibility is tracked over time, not sampled once

- **When** citation in an AI answer is measured
- **Then** it is measured repeatedly and stored as a series
- **Never** quoting a single lucky result as the position

R19.6 A sitemap lists only pages that exist and are meant to be found

- **When** a sitemap is generated
- **Then** it contains live, indexable pages
- **Never** listing archived or blocked pages, which wastes the crawl on nothing

Module 20 · Projects & Collaboration — 9 rules

R20.1 Billable time becomes an invoice line without retyping

- **When** approved time exists against a project
- **Then** the rate card turns it into an invoice line and a real cost
- **Never** re-entering hours into an invoice, which is where the two figures start to differ

R20.2 Billable and non-billable are separated at entry

- **When** time is recorded
- **Then** it is marked billable or not as it is entered
- **Never** deciding at invoice time, which quietly turns unbillable work into a charge

R20.3 An approval shows the rule that demanded it

- **When** anything lands in the approvals queue
- **Then** the rule that sent it there is displayed beside it
- **Never** presenting a request with no stated reason, which makes approval a formality

R20.4 An approval decision goes to the audit trail

- **When** a request is approved or refused
- **Then** the decision, the person and the time are recorded
- **Never** recording only the outcome on the record, which loses who accepted the risk

R20.5 An automation run is kept step by step

- **When** a rule fires
- **Then** what triggered it, each step, and what each step returned are stored
- **Never** keeping only the outcome — an automation nobody can inspect afterwards is a rule the business cannot trust with its money

R20.6 An automation acts within a named scope

- **When** a rule is built
- **Then** the records it may read and write are declared on it
- **Never** letting a rule reach anywhere in the system because it happens to run as an administrator

R20.7 A project cost includes the time and the material

- **When** project profitability is computed
- **Then** labour, material and expenses booked to it are all included
- **Never** reporting on revenue and time alone, which shows a loss-making project as profitable

R20.8 A decision is recorded where the decision was made

- **When** a discussion resolves something
- **Then** it is attached to the record it concerns
- **Never** leaving the reasoning in a chat thread that will not be found in a year

R20.9 A procedure is scoped to the role it applies to

- **When** a standard procedure is published
- **Then** it is scoped to the role that performs it
- **Never** publishing one undifferentiated manual that nobody reads

Module 21 · Dashboard & BI — 9 rules

R21.1 The group figure is the sum minus inter-company trade

- **When** several companies are consolidated
- **Then** entries naming a counterparty inside the group are eliminated, and gross, eliminated and group are all shown
- **Never** presenting the plain sum as the group result, which inflates turnover by trade the group never did with the outside world

R21.2 An entry cannot be its own counterparty

- **When** an entry names a counterparty company
- **Then** it is refused if that is the same company
- **Never** allowing a company to trade with itself, which eliminates a figure that was never doubled

R21.3 The number of companies is data, not a constant

- **When** the group grows

- **Then** a company is a row and every consolidation follows
- **Never** building around a fixed number of companies or channels

R21.4 Every dashboard figure is a live query

- **When** a KPI is displayed
- **Then** it is computed from the ledger and the stock table at that moment
- **Never** reading a maintained summary table, which can be wrong without looking wrong

R21.5 A consolidated row is a formula over the company rows

- **When** a workbook or report shows a consolidated figure
- **Then** it is computed from the company rows beside it
- **Never** typing a separate consolidated total, which is a second copy that will disagree

R21.6 A figure a user may not see is not returned

- **When** a report runs for a scoped user
- **Then** out-of-scope rows are excluded from the query
- **Never** computing the full figure and hiding part of it in the display

R21.7 An exported report says when it was taken

- **When** a report is exported
- **Then** the as-at time and the filters are printed on it
- **Never** producing an undated export, which is quoted months later as though it were current

R21.8 A saved report keeps its definition, not its results

- **When** a report is saved and re-run
- **Then** the definition re-runs against current data
- **Never** storing a snapshot and presenting it as live

R21.9 A figure with no drill-down is a defect

- **When** any total is shown
- **Then** it opens to the records beneath it
- **Never** shipping a number that cannot be explained by clicking it

Module 22 · AI Assistant, Agents & Automation — 15 rules

R22.1 An answer carries the records it came from

- **When** the assistant answers a question about a figure
- **Then** the rows it used are attached and each one opens to its record
- **Never** giving a bare number, which cannot be checked and therefore cannot be trusted

R22.2 An unknown answer is said, never estimated

- **When** the assistant cannot find the figure
- **Then** it says so and shows what it looked at
- **Never** producing a plausible number — a confident wrong figure costs far more than an honest blank

R22.3 The assistant answers only from what the asker may already see

- **When** a question is asked by a scoped user
- **Then** retrieval is filtered to that user's permissions before the answer is composed
- **Never** letting the assistant become a way around permissions that every other screen enforces

R22.4 An agent cannot widen its own scope

- **When** an agent runs
- **Then** it works within the records and the spend it was given
- **Never** expanding its scope mid-run, however sensible the next step would be

R22.5 Money never moves without a human yes

- **When** an agent proposes a refund, a payment, a payout or a credit note
- **Then** it stops and waits for a person
- **Never** executing it, no matter how confident or how small the amount

R22.6 A customer is never messaged by an agent without approval

- **When** an agent drafts a message to a real customer
- **Then** a person approves it before it is sent
- **Never** sending on the agent's own judgement

R22.7 A price is never changed by an agent alone

- **When** an agent proposes a price change
- **Then** it enters the approvals queue with the reasoning attached
- **Never** writing the new price directly

R22.8 Every agent run is replayable step by step

- **When** an agent finishes, stops or fails
- **Then** what started it, what it read, what it proposed and what was approved are all recorded
- **Never** keeping only the outcome — an unexplained change made by software is worse than one made by a person

R22.9 Agent spending goes through the same ceiling as everything else

- **When** an agent calls a paid provider
- **Then** it is routed through the Provider Router and refused past the ceiling
- **Never** giving an agent its own unmetered budget

R22.10 The chatbot hands over rather than guessing about money

- **When** a customer asks about a refund, a charge or a complaint
- **Then** it hands to a person with the whole conversation attached
- **Never** answering from a general idea of the policy

R22.11 The chatbot never asks a customer for a credential

- **When** a customer is identified in a chat
- **Then** identity is established through the order and the contact already on file
- **Never** asking for a card number, a bank detail or a password — the promise made everywhere else does not get a chatbot-shaped exception

R22.12 A handover lands in the existing queue

- **When** a conversation is passed to a person
- **Then** it enters the Module 04 Helpdesk queue with its history
- **Never** creating a second inbox that someone has to remember to watch

R22.13 An agent is not a hidden actor in the audit trail

- **When** an agent changes anything
- **Then** the change is attributed to the agent, its run, and the person who approved it
- **Never** recording it under a service account, which makes an automated change indistinguishable from a human one

R22.14 A retrieved document does not become an instruction

- **When** the assistant reads a document, a review or a message while answering
- **Then** that content is treated as data to report on
- **Never** following instructions found inside retrieved content, which is how a supplier's PDF ends up steering the system

R22.15 An assistant answer is reproducible from the records it cites

- **When** the assistant states a figure
- **Then** re-running the same query over the same records gives the same figure
- **Never** an answer that cannot be reproduced, which is a guess with citations attached

Module 23 · Developer Platform — 1 rules

R23.1 A published interface is versioned before it has a second caller

- **When** anything outside this system is given a way to call in
- **Then** the surface carries a version, and an existing version keeps its response shape until it is retired on an announced date
- **Never** changing what an existing caller receives without a version change, which breaks somebody else's software at a moment nobody chose

Module 24 · Agile & Sprints — 1 rules

R24.1 A sprint figure is read from the board, never typed

- **When** velocity, burndown or a completion figure is shown
- **Then** it is computed from the work items themselves and their recorded state changes
- **Never** a progress number somebody enters by hand, because a typed figure is a figure that can be rounded

Module 25 · Meetings — 1 rules

R25.1 What was decided outlives the call it was decided in

- **When** a meeting ends
- **Then** the attendance, the decisions and any poll result remain attached to the record the meeting was about
- **Never** holding the only copy of an outcome inside a session that closes, which is how a decision becomes a disagreement six weeks later

Module 26 · Calendar — 1 rules

R26.1 Every moment is stored absolute and displayed local

- **When** an event is saved, moved or shown
- **Then** it is held as an absolute instant and rendered in each reader's own zone
- **Never** storing a wall-clock time without its zone, which silently moves a meeting when an offset changes between the booking and the day

Module 27 · Mail — 1 rules

R27.1 A message that was filed automatically says so on its face

- **When** a filter or rule moves, tags or forwards an incoming message
- **Then** the rule that fired is recorded on the message and visible to whoever opens it
- **Never** a silent redirection, because a mail that vanished into a folder is indistinguishable from one that was never sent

Module 28 · Documents — 1 rules

R28.1 A document belongs to the record it is about

- **When** a document, presentation or note is created here
- **Then** it is filed against the order, party, case or project it concerns and is found by that record
- **Never** a document whose only address is a folder somebody has to remember, which is the filing cabinet this module exists to replace

Module 29 · Learning — 1 rules

R29.1 Completion is a dated record against a named person

- **When** somebody finishes a course or passes an assessment

- **Then** the completion is recorded with the person, the material version and the date, and can be produced when an auditor asks
- **Never** reporting that training was offered as though it were training completed, which is the distinction an inspection turns on

Module 30 · Integration Platform — 1 rules

R30.1 A connector runs with the scope it was granted, not the scope it asks for

- **When** an integration reads or writes through the framework
- **Then** it acts within the permissions recorded when it was installed, and widening them is an approval with a record
- **Never** a connector that quietly extends its own reach mid-run, which turns an integration into an unaudited second way in

Module 31 · Identity & IT — 1 rules

R31.1 A second factor is required before the system is reachable from the internet

- **When** sign-in is available anywhere outside a private network
- **Then** a second proof is enforced for every account that can see money, staff or customer data
- **Never** relying on a password alone on a public address, because one reused credential elsewhere then becomes a breach here

M15 · EVERY APP, UNDER ITS MODULE

Not a count — the list. A count tells a reader how much they are not being shown.

31 modules · 165 apps. The whole of what is being built, named. A module is a part of the business; an app is one screen-and-its-work inside it. Any of them can be switched off for a business that does not need it — see the changeable things.

Module 01 · Platform — 9 apps

- **Identity, Settings & Audit** — Users, roles and permissions, company switching, tax and numbering setup — and an immutable record of everything that ever happened.
- **Industry Packs** — What your trade calls things, the stages your work moves through, the extra fields your records need, the documents you issue and the reference data you start with — all of it loaded as a row of configuration, never as a separate version of the software. Pick the pack that matches your trade and the whole system changes its words: the same order screen reads as a job, a matter, a consignment or an appointment, with the same columns underneath. A pack may rename what exists, add fields to tables that exist, and switch discretionary rules off; it may never invent a concept, add a field to a table that does not exist, contain any executable code, or switch off the guarantees — company scoping, the audit trail, money never being a float — because a trade may change its vocabulary and may not opt out of the things the books are

trusted for. Adding a trade nobody anticipated is therefore a file somebody writes, not a release somebody ships.

- **Ask & Print** — Ask from your phone, anywhere: a ledger, a bill, today's packing slips. It comes back as a PDF — or prints on the office printer, with nothing plugged into your phone.
- **Communications** — WhatsApp commands and broadcasts, email and SMS, and the handful of scheduled jobs that carry a nudge to the right person without anyone remembering to send it. Every capability here has more than one interchangeable provider — none of them is ever the source of a figure the business reports.
- **WhatsApp Command Console** — The shop floor does not open a laptop. A worker or a staff member sends a short message — in, out, today's pieces, leave, an advance — and it becomes a real record: attendance with the time and place it was marked, a production report against the design, a request in the approvals queue. The end-of-day prompt asks what is still open and how long it needs, so tomorrow starts from what is actually pending rather than from memory. Marking attendance checks the phone is at the unit within the radius set for it, with a grace window; a person outside it is not refused, they are flagged for the manager, because a system that locks someone out of being paid for being early at the wrong gate has failed at its job. Every override is recorded with who made it. The words a worker types are in the language they speak, and nothing here ever asks for a password, a bank detail or a document number.
- **Data Privacy & Consent** — What a person's data may be used for, captured as consent at the point it is given and honoured everywhere downstream — including the right to have it corrected or removed. Retention (how long a record is kept) and deletion (a person's right to have their own data removed) are two different policies, tracked separately, because a rule that keeps records for the law and a request from a person to be forgotten do not resolve the same way.
- **Provider Router & Cost Guard** — The rule that no capability depends on one outside service, enforced at the moment it matters instead of merely promised. Every capability has an ordered fallback list ending on an option that needs nothing connected, so a courier API that stops answering at 9pm, or an AI key that hits its quota mid-catalogue, drops to the next option instead of stopping the work. A provider that keeps failing is tripped out of the list entirely and retried once after a cooldown rather than hammered; retries inside one provider wait twice as long each time. And every paid call is counted in paise against a ceiling you set — over the ceiling the paid provider is refused, not warned about, and the work completes on a free one. Because every capability is guaranteed a built-in or by-hand option, a spent budget can stop the spending without ever stopping the business.
- **Payment Data Scope** — A written statement of exactly which systems ever see a card or bank credential, and which never do — because every card-capable screen in this system hands that moment to a payment provider's own secured field and never stores or even passes the number through application code. The statement is what an auditor or a partner asks for before they will connect to this system.
- **Appearance & Localisation** — The two settings a person notices in the first minute and nobody writes a specification for: which time zone a screen shows its dates in, and whether it is drawn light or dark. Time zone is the one that matters for correctness rather than comfort — every timestamp in this system is stored as an absolute moment, so a business working across two zones reads the same event at the right local time in both, and a working day that ends at midnight ends at midnight where the work happened.

Module 02 · Design & Sampling — 2 apps

- **PLM & Development** — First idea to something you can actually make: specification, sample rounds, costed trials and sign-off, with every version kept. A product, a machined part, a formulation or a service package all

move through stages you set yourself.

- **Design / IP Register** — What protects a design once it exists — trademark or copyright status, the date it was first shown, and a flag the moment a near-identical listing turns up elsewhere. Every version already kept by PLM & Development gets a filed status here instead of just a date stamp, because a design with no ownership record on file is a design nobody can defend.

Module 03 · Inventory & Catalog — 4 apps

- **Stock** — Live quantity by SKU, location and stage, with reorder alerts, batches, kits and dead-stock. Goods you still own but that sit in a channel's own warehouse are a location like any other, so consignment and sale-or-return stock is counted, valued and aged with everything else instead of disappearing off the books until it sells.
- **Catalog / PIM** — One product record — attributes, images, HSN, MRP and the price each channel actually sells at — pushed to every marketplace and to your own storefront, and scored for each channel's rules before it lists. It also holds the two things everything downstream depends on: the code each channel knows this product by, mapped to yours, and the packed size and weight that decide the courier rate and settle every weight dispute. Every listing's state on every channel is here too — live, waiting for your approval, blocked, archived — with the quality score that decides whether anyone sees it.
- **Kit & Combo SKU** — A sellable SKU made of component SKUs — a three-piece set sold as one listing. Selling the kit decrements each component at order time, so stock is right for every piece in the set, not just the set itself.
- **Master-Data Hygiene** — Duplicate detection and merge across customers, vendors and designs, and a dead-stock register for what has not moved in months. Protects every downstream report from the same record existing twice under two names.

Module 04 · CRM — 5 apps

- **CRM & Customer 360** — Lead to won, then the full lifetime: orders, returns, value and what to offer next.
- **Documents & eSign** — Every agreement, receipt, certificate and scan filed against the record it belongs to — an order, a party, a case, an employee — so it is found by that record instead of by remembering a folder. Send one out for signature and the signed copy files itself back.
- **Helpdesk & Live Chat** — Questions arriving by chat, email or phone become tickets tied to the order or the account they are about, with the whole history already on the screen.
- **Forms & Feedback (NPS)** — A short form after delivery, and the score it produces attached to the design or item it is actually about — not just the buyer — so a complaint-prone item surfaces as a pattern instead of a scatter of individual gripes.
- **Territories** — Dividing customers and targets by region or by team, and reporting against that division. Worth having the moment there is a sales team large enough to divide and an argument about whose account something was; pointless before that. A territory is a row like anything else here, so redrawing the map is an edit rather than a migration, and last year's figures stay readable against last year's boundaries instead of being silently restated under the new ones.

Module 05 · Sales — 9 apps

- **D2C Sales** — Orders from your own storefront — hosted, self-hosted or custom — cart to dispatch, with loyalty and partial COD.
- **B2B & Credit** — Wholesale orders with credit limits, tier pricing and outstanding ageing.
- **Export** — Commercial invoice, packing list, LUT bond and IGST-refund tracking.
- **POS** — Counter billing that draws on the same stock as your website.
- **Quotes & Proforma** — Send a quote, convert it to a confirmed order in one click.
- **Couriers & AWB** — Book the shipment on the order itself, compare couriers, print the label and follow the AWB to the door.
- **Subscriptions** — A schedule that raises its own invoice on its cycle and follows up on its own when a payment fails — for anything sold as a standing order rather than a one-off.
- **Customisation & Made-to-Measure** — The order that does not exist in the catalogue: a buyer sends reference pictures and their own measurements, a price is agreed over several messages, and the piece is made for them. All of it is one record — the references, the measurement set, every quote in the negotiation and what was finally agreed, the advance taken to start work and the balance taken before dispatch. When the order is accepted it opens a production order like any other, so a bespoke piece is costed, made, checked and posted exactly as a catalogue piece is. Two legs of money on one order is the part most systems get wrong: the advance is earned when the work starts, the balance is owed until the piece ships, and the ledger shows both separately rather than one payment appearing when the whole thing is over.
- **Appointment Booking** — A slot a customer books for themselves — a fitting, a consultation, a service visit — against the real availability of whoever or whatever is needed for it. It writes to the same Calendar the rest of the business works from, so a booking taken on a website at midnight cannot collide with something already in the diary.

Module 06 · Planning & Requirements (MRP) — 3 apps

- **Demand Forecast & Signal** — What sold, by SKU and by period, turned into a short-term forecast — the number every requisition and every production order downstream is measured against instead of a guess.
- **Requirement Explosion (MRP run)** — Confirmed demand exploded through the bill of materials into what raw material to buy and what to produce, and by when — so a purchase requisition or a production order always traces back to real demand, never to a feeling that stock is getting low.
- **Open-to-Buy / Budget Ceiling** — A spending ceiling set per period regardless of what the demand signal suggests, so a real signal cannot on its own commit more money than the business has decided to risk this season. Every requisition the MRP run drafts is checked against what is left of the ceiling before it goes to Purchase.

Module 07 · Purchase — 3 apps

- **Procurement** — RFQ to purchase order to goods receipt, with a strict three-way match before any bill is paid.
- **Vendor Management** — Vendor 360, payables, ageing, a real risk score, and sourcing that follows performance.
- **Insurance Register** — What cover exists over stock in transit, stock in the warehouse and product liability, matched against what is actually moving through Purchase and Warehouse right now — so real value in

transit is never quietly running with no cover tracked anywhere.

Module 08 · Manufacturing — 4 apps

- **Production Orders** — Your own stages from first operation to finished goods, with work-in-progress visible at each one.
- **Piece-rate & Contractors** — Output-based pay for anyone paid by the piece rather than the hour — pooled completion, per-unit rates, rework and advances resolved into a single payout.
- **BOM & Consumption** — What each product consumes, costed at today's material rates.
- **Maintenance** — Machines, tools and assets: what is due for service, when it was last done, what it cost, and what stopped while it was down. Planned and breakdown work against the same asset record.

Module 09 · Quality & Compliance — 2 apps

- **Quality Control** — Accept, reject or rework — on goods received and on goods made — with reasons that feed the supplier scorecard and the performance flags alike.
- **Certificate & Compliance Register** — Every standard the business or a vendor holds — a safety, labour or environmental certification — with its issue date, its expiry, and the audit that backs it, so a certificate about to lapse is visible before a buyer asks for it and finds it already expired. What this register tracks is what a sustainability report downstream is actually built from.

Module 10 · Warehouse — 3 apps

- **Picking & Bins** — Pick lists that tell staff exactly which bin to walk to, in walking order.
- **Barcode Operations** — Scan to pick, pack, dispatch and run a physical stock count from a phone — the same scan whether the order came from a marketplace, your storefront site or the counter.
- **Packing Video** — Every parcel recorded as it is packed and indexed by its order number, so a wrong-item claim is answered with the clip. The footage attaches itself to the claim that needs it.

Module 11 · Logistics — 5 apps

- **Rates & Zones** — Every courier's rate card by zone, weight slab and service — so the cheapest and the fastest option for this parcel are both known before it is booked.
- **NDR & RTO Rescue** — A failed delivery worked while it can still be saved — reattempt, call, correct the address — before it becomes a return you pay for twice.
- **COD Remittance** — What the courier collected at the door against what reached your bank, parcel by parcel, with every shortfall named and aged.
- **Handover & Manifest** — What is expected out today against what the courier actually took, counted per courier and per service. The manifest to hand over, the one-time code to confirm it, and a signed record of the parcels that were left behind — so a parcel lost between your table and their van has an owner.
- **Fleet** — Your own delivery vehicles, if you run any — what each trip cost, what is due for service, and what that adds to freight. Optional; most businesses run couriers only.

Module 12 · Accounting & GST — 9 apps

- **Accounting** — Double-entry books where every voucher balances and the trial balance always ties.
- **Invoicing** — GST tax invoices and receipts, totals computed from the lines to the paise. Where a channel raises its own invoice, both numbers live on the order — theirs and yours — so the panel's paperwork and your books point at the same sale and neither has to be re-keyed to find the other.
- **Expenses** — Spend captured by category with approvals, and bill OCR to save typing.
- **GST & Tax** — CGST, SGST, IGST, TDS, TCS, input credit, GSTR-1 and GSTR-3B — filed per registration, so a group where one company is registered and another is not files exactly what each one owes and nothing it does not.
- **ITC Reconciliation** — Every input credit matched against the government's own GSTR-2A/2B before a return is filed, so a credit claimed is a credit that is actually on record.
- **Receivables, Payables & PDC** — Payments and receipts allocated against named open invoices, and a register for post-dated cheques that posts only on the date they are realised, not the date they were written.
- **Fixed Assets & Depreciation** — The asset register with both Straight-Line and Written-Down-Value depreciation tracked side by side, and disposal posting its gain or loss straight to the books.
- **Year-End Close & Period Lock** — Profit-and-loss accounts reset and balance-sheet accounts carry forward at year end, and a reviewed period locks against backdated edits until an admin unlocks it — an act that is itself logged.
- **Finance Reports** — P&L, balance sheet, and profit by channel, product and SKU.

Module 13 · Treasury & Financial Planning — 3 apps

- **Cash Flow Forecast** — Expected receipts and expected payments laid out by week, drawn from open invoices and open bills rather than typed in by hand — so a cash shortfall is visible weeks before the date it would actually bite.
- **Banking & Reconciliation** — Bank statement lines matched against the ledger's own record of what should have moved, with anything unmatched surfaced instead of silently carried forward.
- **Budget vs Actual** — A budget set per category and period, and actual spend from Accounting tracked against it as the period runs — not only after it closes, when the only thing left to do is explain the variance.

Module 14 · Settlement — 3 apps

- **Payout Cycles** — Every settlement cycle each panel runs — what it should pay, what actually landed in the bank, and on which day — so a late payout is visible the day it is late, not at month end.
- **Fee & Commission Audit** — The rate card a channel publishes against the rate it actually charged, category by category and SKU by SKU. A silent commission increase is caught the first time it is applied — and the tier you are rated in is on the same screen, because the tier is what the rate card hangs off, and losing one quietly costs more than any single deduction.
- **TCS & TDS Register** — Every rupee the panels deducted as TCS and TDS, matched against the portal's own figures — so the credit you claim is the credit you are actually owed.

Module 15 · E-commerce / OMS — 11 apps

- **Marketplace OMS** — Every marketplace and every storefront in one order queue — Amazon, Flipkart, Myntra, Meesho, Ajio, Nykaa and JioMart alongside any hosted, self-hosted or custom storefront. The stages each channel really uses — to accept, to pack, ready to dispatch, handed over, in transit — with the right cut-off counting down on every order, because a quick-commerce or air-shipped order is not due at the same hour as a standard one. Priority orders first, the day grouped by product so one item is picked once instead of once per parcel.
- **Order Management** — One pipeline from new to delivered, whether the order came from a seller panel, your own storefront site, a dealer or the counter.
- **Manual Data Check** — Upload the sheets you already download — marketplace orders and returns, and your own counter-shop registers, one file or a whole ZIP — and read ten cross-checks back: money, month, item, state, returns, claims, ads, payouts and GST. Every figure is clickable down to the transactions behind it, and the whole result downloads as Excel.
- **Reconciliation** — Match every marketplace payout to the order line that earned it, and expose the gap.
- **Claims & Disputes** — Turn shortfalls, weight disputes and lost parcels into filed claims with evidence — and answer them before the clock runs out. A claim that is awaiting your response is worth money; one closed for no response is worth nothing, so the days remaining sit on the screen next to the amount.
- **Returns / RMA** — Customer, courier and wrong returns — and the dead stock they actually cost you.
- **Channels & Storefronts** — Connect a channel once and it stays in step: catalogue out, price out, stock out, orders in. Seven marketplaces and any website platform — any hosted, self-hosted or custom site over its own API — each switchable without touching your data. Shopsy and any other storefront a channel runs alongside its main one counts as its own channel here. Where a channel has no open interface, its own downloaded report is a first-class way in. A channel may also know you by a different trading name — that is a label on the channel, not a second company, so it tags the order and the payout without ever splitting your books.
- **Labels & Documents** — The channel gives you a PDF; this turns it into something a packer can work from. Cropped to your label size, your own product code printed large where the channel left it off, the invoice and the packing slip merged behind it, and the whole batch sent to the label printer in one job. Reprint a single parcel without redoing the batch — and nothing is ever uploaded to an outside website to be cropped.
- **Listing & Catalog Manager** — Bulk-create and bulk-edit listings across every channel from the one product record in Inventory & Catalog, and catch the mismatches that quietly cost sales: listed but out of stock, or in stock but never listed.
- **Size / Fit Recommendation AI** — A fit suggestion at the point of purchase, built from the item's own measurements and the return history of buyers who picked each size — aimed straight at the return reason that costs the most: the right item in the wrong size.
- **AR / Virtual Try-On** — A way to see drape, fit and colour on a screen before buying, for items where a flat photo alone leaves too much to guess.

Module 16 · HR & Payroll — 6 apps

- **Staff & Contractors** — Attendance, effective-dated salary and output-based earnings in a single register, whoever is on it.
- **Time-off & Advances** — Leave, festival advances, and exactly how they change this month's payout.

- **Appraisal & Hiring** — Performance reviews and a hiring pipeline that ends in an employee record.
- **Recruitment** — The pipeline before someone becomes an employee — an opening, the people who applied for it, a trial piece where the work itself is the interview, and the decision with its reason kept. It matters more in a skilled trade than in most: a person is taken on for skill at one particular kind of work, and the trial output is the evidence, so it is recorded against that work and the rate that would apply rather than remembered as an impression. A candidate who is not taken on now stays findable when the same skill is needed in a busy month, and their personal documents are held under the same consent and retention rules as anyone else's, not in a folder on somebody's phone.
- **Payout Execution** — Where the calculation in the earnings register actually turns into money leaving the business — bank batch, UPI, cash against a signed receipt — with the method and the reference recorded against every payout, so the register's total and the money that actually moved can always be checked against each other.
- **Employee Engagement & Surveys** — Asking the people who work here how it is going, and keeping the answers where a pattern can be seen rather than in one manager's memory. Anonymity has to be real to be worth anything: a survey that quietly identifies its respondents produces polite answers and no information, so where a response is promised anonymous it is stored unlinked from the person, and the system cannot re-join them afterwards even for an administrator.

Module 17 · Marketing — 9 apps

- **Social Calendar** — Plan and publish across every channel from one calendar.
- **Campaigns** — Email, SMS and WhatsApp campaigns measured on real revenue, not opens.
- **Repricing Engine** — Rules per channel and SKU — floor, ceiling, match-lowest, festival overrides — and what each change actually did. A price that went up and took the orders down with it shows as exactly that, next to the rule that raised it, so the rule can be reversed on evidence rather than on a feeling.
- **Automation** — If this happens, do that — across any module, without writing code.
- **Blog & Pages** — Articles, landing pages and category copy written, scheduled and published straight to your own site — any hosted, self-hosted or custom storefront — with the meta title, description and internal links set before it goes out.
- **Events** — Trade shows and exhibitions worked as a channel of their own — booth, budget and every lead captured on the floor landing straight in CRM instead of on a stack of business cards.
- **Website & Page Builder** — The storefront itself, built by dragging sections into place rather than by editing a theme file — hero, product grid, size guide, lookbook, contact form — each block reading live from the catalogue, so a price or a stock state on a landing page is the same number the order screen uses instead of a figure someone pasted in and forgot. Blog & Pages above writes articles into a site that already exists; this is for the businesses that do not have one, and it is the gap that shows up plainly when this module list is set beside a mature open-source ERP: they ship a full site builder next to the blog, and until now this did not.
- **Markdown / Clearance Optimization** — The same rule engine that reprices for competitiveness, aimed at ageing stock instead: when to start discounting it and by how much, before it becomes a warehouse write-off rather than a sale at a lower margin.
- **Social Engagement & Inbox** — The replies, comments and messages that arrive on the accounts the business posts from, in one queue rather than four apps on somebody's phone. Distinct from the Social Calendar, which decides what goes out: this is what comes back, and it is where a complaint first appears in

public. A message here can become a Helpdesk ticket with the thread already attached, so a public question and a private one end up in the same place.

Module 18 · AI Content Engine — 8 apps

- **Content Engine** — Fourteen stages in your own voice, from buyer research and competitor reading through hooks, channel-ready listings, ads, social posts, video scripts, song lyrics, the publishing calendar and alt text — each written from your own catalogue, so the words match the thing.
- **Image Studio** — Layers, free transform, background removal, channel presets and SEO alt text — a phone photo becomes a channel-compliant product image.
- **Video Studio** — Text and image to video, reels and ad cuts sized for every channel.
- **Design Studio** — A full design surface — templates, layers, undo and redo, any colour, exact sizing, background images and stock elements — exporting PNG, JPG or PDF at whatever size the channel or the printer asks for.
- **Motion Renderer** — A reel rendered from a page of HTML and CSS — the same layers, fonts and brand colours the Design Studio already uses — into a real MP4, on this machine, with nothing uploaded. It is not a screen recording: the clock is faked, the animation is seeked to the exact instant of each frame, and only then is that frame captured, so the render does not care whether the machine was busy. Rendering the same festival banner twice produces the same file to the byte, which is what makes a reel something you can check and re-cut rather than something you have to watch all the way through and hope about.
- **Narration Studio** — A voice over the reel, in the language the buyer actually speaks — the same script the Content Engine wrote, spoken. The default needs nothing installed and no key, because every modern browser can already speak; a self-hosted or cloud voice sits behind it as an interchangeable provider for anyone who wants a cloned or branded one. Long scripts are split at sentence boundaries and rejoined, so a two-minute description is not cut off at whatever limit a service imposes. A voice cloned from a real person is only ever used with that person's recorded consent, filed against them in Data Privacy & Consent like any other permission.
- **Image Generation Slot** — Generated imagery — a model on a background you do not have to shoot, a festival backdrop, a lifestyle scene — as a capability with interchangeable providers rather than a bet on one service: a queue of jobs, a preview while it works, inpainting to fix one region, upscaling and face correction. This one needs a graphics card. Image models cannot run on an ordinary office machine or on the container this system is built in, so what ships is the queue, the review screen and the provider slot, with the generating itself done by whichever engine you point it at — your own GPU box, or a cloud service, swapped without touching anything else. Said plainly here because the alternative is a screen that looks finished and produces nothing.
- **Publisher** — One push sends the finished listing, picture and copy everywhere it has to appear — your storefront, each marketplace, each social account — and reports back what actually went live and what was rejected, with the reason.

Module 19 · SEO, AEO & AIO — 3 apps

- **Technical SEO & Schema** — Structured data, sitemaps and page-level technical checks against your own storefront and content pages, so a search engine can read what a page is actually about instead of guessing from the text alone.

- **Answer-Engine Optimization** — Content shaped to be quoted directly by an answer box or a voice assistant — a clear, citable answer near the top of the page — rather than written only for a person to scroll through.
- **AI-Engine Visibility Tracking** — Whether and how this business is actually cited when someone asks an AI assistant a shopping question in this category, tracked over time — the same discipline as rank tracking, aimed at a newer kind of result page.

Module 20 · Projects & Collaboration — 7 apps

- **Projects & Cases** — A project, a case file, an engagement or a job — whatever your work is called. Stages you define, owners, deadlines, documents, billable time and real cost, all on one record the ledger can see.
- **Timesheets & Planning** — Who is on what this week, and the hours that actually went in — against a project, a case, a job or a machine. Billable and non-billable separated, so a rate card turns straight into an invoice and a real cost.
- **Approvals** — One queue for everything waiting on a yes — a purchase order, a discount, a leave day, a credit note, a payment. The rule that sent it there is on the screen next to it, and the decision goes to the audit record.
- **Forum** — Questions and answers that outlive a chat — for customers, dealers or staff — with the useful ones kept where the next person will actually find them.
- **Automation Studio** — The place a person builds “when this happens, do that” by dragging it out and watching it run — a trigger, the steps after it, a branch where the answer decides which way to go — over the same event stream every module already writes to. Marketing’s Automation aims that idea at campaigns; this is the general one, reaching any module: a payment marked short holds the next dispatch and opens a claim, a worker’s pooled output crossing a threshold raises the payout for approval, a return marked damaged writes off the piece and messages the buyer. Every run is kept — what fired it, each step, what each step returned — because an automation nobody can inspect afterwards is a rule the business cannot trust with its money.
- **Discuss** — Conversation attached to the record it is about: this order, this bill, this case. A year later the reason for a decision is still sitting next to the decision.
- **Knowledge Base** — A searchable internal wiki of standard operating procedures, scoped to the role it applies to, so how a task is meant to be done is written down once instead of carried in one person’s head.

Module 21 · Dashboard & BI — 9 apps

- **CEO Dashboard** — Cash, sales, stock, profit and alerts on one screen, refreshed as work happens.
- **Report Builder** — Drag the fields you want into a report and save it for the whole team.
- **Group Consolidation** — Several companies, one set of figures — sales, cash, stock and profit rolled up across every company you run, inter-company entries removed, with years of history to compare against. Add a company whenever the business grows one; nothing in the software caps the number, only the plan does. And a company with no tax registration of its own — a job-work arm, a new venture not yet registered — is a company like any other here, kept in the group figures without being dragged into a return it does not belong in.
- **Excel Dashboard Builder** — A full workbook — financial summary, HR, purchase, sales, inventory and production, GST, expenses — generated from the live records behind every other screen, with each company shown as its own row and a consolidated row that is a formula over them, never a separately typed total.

- **ESG / Sustainability Reporting** — Water usage, chemical compliance, waste and packaging, reported from the same certificate and audit records Quality & Compliance already keeps — so a sustainability report is a query over evidence already on file, not a separate exercise assembled once a year from scratch.
- **Data Warehouse** — Somewhere shaped for asking questions rather than for recording transactions. Honestly premature: a warehouse is what a business builds when queries against the live records start hurting, and nothing here has ever run at a volume that could hurt. It is named so the specification has a home for it, and ranked as what it is.
- **Pipelines & Enrichment** — The scheduled movement of data into that shape, and the step that fills in what a record was missing from somewhere else. Enrichment is the part that needs care: a field filled in from an outside source has to stay visibly different from one the business entered itself, or a guess quietly becomes a fact in next quarter's report.
- **Maps & Geospatial Views** — Figures drawn on a map instead of in a column, for the questions that are actually about geography — where the orders come from, where the returns concentrate, which routes cost what. One visualisation type among many, listed separately in the specification and kept separate here because it needs location on the records before it can show anything at all.
- **Embedded Analytics** — A report placed inside another screen, or inside somebody else's system, rather than opened from the reports list. It reads through the same permissions as everything else, so an embedded view can never show a viewer more than they could have seen by opening the report directly — which is the mistake that makes embedding dangerous when it is bolted on afterwards.

Module 22 · AI Assistant, Agents & Automation — 7 apps

- **AI Assistant** — Ask in your own words — “what did Myntra actually pay us last week, and what is still short?” — and get the answer with the rows it came from sitting underneath it, each one clicking through to the record. It reads the ledger, the stock table and the settlement lines the same way a report does, so the figure it gives is the figure the books give. When it cannot find the answer it says so and shows what it looked at; it never estimates a number and presents it as a fact, because a plausible wrong figure is far more expensive than an honest blank. It answers only from records the person asking is already allowed to open, so it can never become a way around permissions.
- **AI Chatbot** — The same engine turned to face the customer, on your own storefront and on WhatsApp: where is my order, will this size fit me, I want to return this. It reads the real order and the real size chart rather than a script written six months ago, and it will say “let me get someone” instead of guessing at anything about money, a refund or a complaint. The handover goes into the Module 04 Helpdesk queue with the whole conversation already attached, so the person picking it up starts where the customer left off instead of asking them to explain again. It never asks a customer for a card number, a bank detail or a password — that promise does not get a chatbot-shaped exception.
- **AI Agents** — A job rather than a question: “chase every unreconciled settlement line from last week and draft the claim for each.” The agent works out the steps, does them, and stops at the point where a person has to decide. It runs inside a scope you set — which records it may read, which it may write, and how much it may spend through the Module 01 Provider Router — and it cannot quietly widen that scope mid-run. Anything that moves money, files a claim, changes a price or sends a customer a message waits for a human yes.
- **Agent Guardrails & Run Log** — What each agent is allowed to touch, written down as a scope rather than trusted to a prompt, and every run recorded step by step: what started it, what it read, what it proposed, what a person approved, what it actually changed. Kept in the same audit trail as everything else, with the same

absence of an off switch. An agent whose working nobody can inspect afterwards is not a colleague, it is an unexplained entry in the books.

- **Knowledge & Retrieval** — The index that makes the answers grounded: your own designs, rate cards, settlement files, standard procedures and past decisions, searchable so a reply quotes what is actually on file instead of what a model remembers about the trade in general. Permission-scoped at the row, so two people asking the same question get answers drawn only from what each of them may already see.
- **Voice In and Out** — The same assistant asked out loud and answering out loud, because the people who most need a figure are often holding something in both hands. A question spoken in the language the speaker actually uses becomes the same query the typed assistant would have run, and the answer comes back as speech with the rows still attached for anyone who wants to look at them. It refuses exactly what the typed one refuses: it will not read out a figure it could not find, and it will not take a password, a card number or a bank detail by voice — a spoken channel does not get an exception to a promise the rest of the system keeps.
- **Corrections & What It Has Learned** — When the assistant gets something wrong, the correction is a record rather than a conversation that scrolls away. Somebody says what the right answer was, that sits against the question it belongs to, and the next person asking gets the corrected version with a note of who corrected it and when. The whole list is readable and editable by the business — not a hidden memory quietly steering answers — so a correction made in haste can be found and undone. Nothing here changes the model; it changes what the model is shown, which is the only kind of learning anybody can audit afterwards.

Module 23 · Developer Platform — 9 apps

- **API & Access Keys** — A versioned surface over the records this system already keeps, with a key per caller, a rate limit per key, and a log of what each one asked for. Versioned from the first release rather than the third, because the moment an outside system depends on a response shape, changing it silently breaks somebody else's software at a time nobody chose. A key is scoped to exactly the records its holder may already read, so this can never become a way around the permissions every screen obeys.
- **Webhooks** — The other direction: this system calling out when something happens rather than waiting to be asked. A subscription names the event and the address to post to, deliveries are signed so the receiver can tell a real call from a forged one, and a failed delivery is retried on a widening interval and then parked where somebody can see it instead of being dropped. Every attempt is recorded, because an integration that silently stopped firing is worse than one that never worked at all.
- **Functions & Scripting** — Small pieces of logic that run inside the system — when a record changes, on a schedule, or when an automation calls them. Each runs with a time limit, a memory limit and the permissions of whoever or whatever triggered it, so a script can never read what its author could not. What a function did lands in the same audit trail as everything else.
- **Low-code App Builder** — Assemble a screen and the record behind it by dragging fields rather than writing code, for the cases a trade needs and nobody anticipated. What it produces is a configuration row, exactly as an industry pack is, and never a separate copy of the software — so an assembled screen still obeys company scoping, the audit trail and every guarantee the rest of the system makes.
- **Builder Data Model** — The tables and relationships a built application stands on: what a record is, which other records it points at, and which fields may not be empty. Kept separate from the builder because the data model is the expensive half to change later — a screen can be redrawn in an afternoon, a relationship that was wrong has to be migrated with all the rows already sitting in it.

- **App Versioning & Release** — A built application has versions, somewhere to try one before it goes live, and a way back. Without this the person who assembled a screen is editing the live system at the exact moment they are still learning it, which is how a business loses a day of entries.
- **Developer Console** — One place to see what the platform is doing for a given caller: keys issued, calls made, webhooks delivered or failed, functions run, and what each one cost. Somebody debugging an integration at two in the morning should not have to ask a person for a log.
- **App Marketplace** — A directory of applications built on this platform that a business can install into its own company, with what each one may read and write shown before installation rather than discovered after it. Nothing installs without somebody first seeing the scope it is asking for.
- **Robotic Process Automation** — Driving another system that offers no interface of its own by working its screens the way a person would. It is the least reliable way to connect anything, and worth having only when the alternative is somebody retyping figures every morning — so it is recorded here as a last resort with a run log proving what it actually did, never presented as equivalent to a real connection.

Module 24 · Agile & Sprints — 4 apps

- **Product Backlog & Stories** — The queue of work that has not been started, written as something a person wants rather than a task somebody was handed, grouped under the larger outcome it belongs to and sized relative to the others. Sizing is deliberately relative and not in hours, because the one thing teams estimate consistently is whether a piece is bigger than another piece.
- **Sprints & Boards** — A fixed period with a set of work committed to it, and a board showing where each piece has got to. The boundary is the point: work added midway is visible as having been added midway, so the difference between what was planned and what was delivered stays legible instead of being quietly absorbed. Planning at the start and a review at the end are part of the same record, not meetings held somewhere else.
- **Velocity & Burndown** — How much this team actually finishes per period, and how the current one is tracking against what remains. Both are read from the board rather than typed by anybody, which matters because a progress figure a person types is a figure a person can round.
- **Release Planning** — Which finished work goes out together and when, drawn from what the sprints have actually completed rather than from what was hoped for at the start. A release date derived from delivered work is the only kind that survives contact with the second sprint.

Module 25 · Meetings — 4 apps

- **Meeting Scheduling** — Proposing a time, holding the slot, and letting people say whether it works before it is fixed. It reads the same availability the Calendar module keeps, so a meeting cannot be booked into a time the system already knows is taken.
- **Participants, Co-hosts & Moderators** — Who is invited, who may run the session, and who may quiet it down. Three separate roles rather than one, because the person who owns the meeting is often not the person who should be managing a room of forty people while also presenting to them.
- **Polls & Q&A** — A question put to everyone with the answers counted, and a queue of questions from the floor that can be ordered by how many people want the same one asked. Both outlive the call: the result is attached to the meeting record rather than disappearing when the window closes.
- **Call & Meeting Analytics** — The record of contact — that a call or a meeting happened, with whom, about which account, and what came out of it — filed against the customer or the project it concerned. This is the

half with lasting value: six months later nobody needs the recording, they need to know what was agreed and by whom.

Module 26 · Calendar — 4 apps

- **Events & Recurrence** — A dated entry with the people it concerns, and the rules for one that repeats. A single occurrence of a repeating event can be moved or cancelled without disturbing the rest of the series, which sounds small and is the difference between a calendar people trust and one they quietly stop using.
- **Invitations, Availability & Time Zones** — Asking somebody to an event, tracking whether they answered, and showing when they are genuinely free. Every time is stored as an absolute moment and displayed in the reader's own zone, so a meeting set across two countries means the same instant to both of them and survives a daylight-saving change landing between the booking and the day.
- **Shared & Team Calendars** — A calendar belonging to a group rather than a person, with control over who may see a full entry and who may see only that the time is taken. The distinction matters: a team usually needs to know somebody is busy without being told what about.
- **Resource Booking** — The things that get double-booked as often as people do — a room, a vehicle, a machine, a piece of equipment. Held as a bookable record with the same conflict checking an invitation gets, because a room with two meetings in it is the same failure as a person with two.

Module 27 · Mail — 5 apps

- **Shared Inbox** — One mailbox several people work, with it visible who is dealing with what so two of them do not answer the same message differently within a minute of each other. An address a business publishes belongs to the business, not to whoever happens to have the password for it.
- **Folders, Labels & Search** — Filing, tagging and finding, including inside what is attached. Search is listed last in the specification and is the one people actually use all day, which is worth saying out loud when deciding what to build first.
- **Filters & Rules** — What happens to a message automatically on arrival: filed, tagged, sent on to somebody else, or pulled to the top. Every rule that fired is visible on the message, because a mail that quietly vanished into a folder somebody forgot they created is indistinguishable from a mail that was never sent.
- **Signatures & Auto-reply** — The block appended to what goes out, and the answer sent while somebody is away. The away message names who to contact instead, so an absence routes the sender somewhere rather than simply telling them to wait.
- **Mail Admin Controls** — Who may open which mailbox, what is retained and for how long, and what may leave the organisation. This is the half that makes a shared mailbox safe to use for anything a business would mind losing.

Module 28 · Documents — 5 apps

- **Word Processor** — Writing a document inside the system so it is filed against the record it concerns instead of living on somebody's laptop. The value is not the editor, which is a solved and crowded problem — it is that the document is attached to the order, the case or the person it is actually about.
- **Presentations** — Slides, for the same reason and with the same honest caveat: the point is where the file lives and what it is linked to, not that this is a better tool than the one already on every desk.

- **Notes** — Short-form writing that is not a document — a thought against a customer, a decision against a project. The format most used and least specified, which is usually a sign it should be built before the two above it.
- **Co-editing & Track Changes** — Two people in the same document without one of them overwriting the other, and a visible record of who changed what so an edit can be accepted or undone. This is the genuinely hard part of all three applications above, and the part a business actually notices missing.
- **Version Recovery** — Getting back what a document said before somebody changed it. Listed once in the specification under storage and worth its own line here, because it is the feature nobody asks for until the first afternoon it is needed.

Module 29 · Learning — 3 apps

- **Courses & Lessons** — The material somebody has to work through, in the order it makes sense, in the language the person being taught actually reads. Built over the staff records that already exist rather than a second list of people maintained beside them.
- **Assessments** — A check that something was understood, at the end of a course or as part of hiring. The same mechanism serves both, which is why the specification lists it twice and this design does not build it twice.
- **Learning Paths & Progress** — A sequence of courses for a role, and who is where in it. The record that matters for compliance is this one: not that training was offered, but that a named person completed it on a date, and that the record can be produced when somebody asks for it.

Module 30 · Integration Platform — 3 apps

- **Triggers & Actions** — The two halves of any integration: something that starts it, and something it then does. Kept as one app because they are meaningless apart — the whole design is that any trigger can drive any action without either knowing what the other is.
- **Connector Library** — A described connection to a named outside system, with how it proves who it is, how often it may be called and its quirks written down once instead of rediscovered by each person who integrates it. A connector nobody can run without credentials still has value as a specification of what connecting would take.
- **Integration Webhooks & APIs** — The framework's own way in and out, distinct from the product surface in the Developer Platform: this is how a connector receives a push from the far side and how it calls back. Separated because the two have different audiences, and conflating them is how an internal mechanism accidentally becomes a public promise.

Module 31 · Identity & IT — 4 apps

- **Second Proof at Sign-in** — A second proof at sign-in beyond a password, with recovery codes for the phone that gets lost and a policy for who must use it. The smallest item in this module and the only genuinely urgent one: a system reachable from the internet with a single factor is a system waiting for one reused password to matter.
- **Password & Secret Management** — Where shared credentials live so they stop living in a message thread, with who may retrieve which one recorded. This system never asks for a marketplace, bank or account

password of its own; this app is about the credentials a business already has to hold for other things, and the fact that they are currently held badly.

- **IT Service Desk** — Internal requests — a laptop, an account, access to something — as tickets with an owner and a resolution. The same shape as the customer Helpdesk in CRM, pointed inward, and honestly a different product from the one this system is for.
- **Security Analytics** — Sign-ins, failures and unusual patterns, read from the audit trail that already records everything. It needs no new capture because the record is already being kept — what is missing is anybody looking at it, which is what this app would be.

M16 · EVERY LAYER, AND WHAT REPLACES IT

Rule 1 in full. M5 says what the stack costs; this says what it is, and — the part that matters — what takes its place. Every layer names what it is built on today, at least two named replacements, and the one interface the rest of the code talks to. That interface is what makes a swap a settings change instead of a rewrite.

`brand/site/checkstack.js` refuses a layer with fewer than two alternatives, a vague alternative ("something else", "any other tool") or no interface — so this cannot rot into a paragraph nobody kept.

19 layers · 57 named replacements. No capability here depends on one company staying in business, keeping its prices or keeping its terms. Each layer names what it is built on today, what can take its place, and the one interface the rest of the code talks to — that last part is what makes a swap a settings change instead of a rewrite.

A check refuses any layer with fewer than two alternatives, a vague alternative ("something else", "any other tool") or no interface, so this cannot rot into a paragraph nobody kept.

The database — PostgreSQL

What it does. Keeps every record — customers, orders, stock, vouchers — and answers questions about them.

Why this one. PostgreSQL is open source, runs anywhere, and has the two things this design needs built in: locks at the record level so one business cannot read another's rows, and exact whole-number arithmetic so money never drifts. Any managed Postgres service is a hosting decision, not a database decision — the same schema runs on all of them.

What can replace it

- A managed Postgres service — same database, somebody else runs the machine
- Postgres on your own server — the software is free, you supply the machine
- MySQL or MariaDB, if a team already knows them well — the schema needs rework for the record-level locks

The rest of the code only ever talks to `DatabaseService` — so changing the line above changes one file, not the application.

What the move actually costs. Moving between Postgres hosts is a dump and a restore. Moving off Postgres entirely means rewriting the isolation layer, which is the one part worth not moving.

File storage — Any S3-compatible object store

What it does. Keeps photographs, invoices and scanned documents — the things too big to sit in the database.

Why this one. Almost every file service speaks the same request format, so one adapter reaches most of them. That makes this the cheapest layer in the whole system to change your mind about.

What can replace it

- A different S3-compatible provider — usually a URL and a key change
- Files on your own server's disk, with a backup copy elsewhere
- A self-hosted object store such as MinIO, which speaks the same format

The rest of the code only ever talks to `FileStore` — so changing the line above changes one file, not the application.

What the move actually costs. Copy the files across and change the address. Nothing above this layer notices.

Cache and short-term memory — Redis, or a Redis-compatible store

What it does. Holds recently used answers and sign-in sessions so common screens open instantly.

Why this one. Nothing here is the only copy of anything. If the cache is wiped the system simply asks the database again and is a little slower for a minute — so this layer can be replaced, restarted or removed entirely without risking a single record.

What can replace it

- Valkey — the open-source continuation of the same thing, same commands
- Memory inside the application itself, which is enough until traffic grows
- A database table, slower but with nothing extra to run

The rest of the code only ever talks to `CacheService` — so changing the line above changes one file, not the application.

What the move actually costs. Near zero by design. Losing the cache loses no data, which is the whole reason it is safe to change.

The backend runtime — Node.js with TypeScript

What it does. Runs the business rules, checks permissions, writes records and calculates totals.

Why this one. The same language runs on the browser side, so one team can work across the whole system and code that validates a form can be shared with the code that validates the saved record — no rule gets written twice and no two versions of it drift apart.

What can replace it

- Any container host — the code is ordinary and carries no host-specific parts
- Python or Go for a service that genuinely suits them, talking over the same API
- A different Node framework — the business logic sits outside the framework on purpose

The rest of the code only ever talks to the HTTP API contract — so changing the line above changes one file, not the application.

What the move actually costs. Low, because the rules live in plain functions rather than inside a framework. Moving a service means moving the functions and putting a different door in front of them.

The API — REST over HTTPS, with a written schema

What it does. The agreed way the screens, the mobile view and any outside system ask the backend for things.

Why this one. Ordinary web requests over predictable addresses. Anything can call it — a browser, a phone, a spreadsheet, another company's software — without a special library.

What can replace it

- GraphQL for read-heavy screens, over the same underlying services
- A direct connection for live screens that must update by themselves
- Scheduled file exchange for partners who cannot call an API at all

The rest of the code only ever talks to the published API schema — so changing the line above changes one file, not the application.

What the move actually costs. Adding a second style is additive — the services underneath do not change.

The frontend — React with TypeScript, screens generated from configuration

What it does. Everything a person sees and clicks — screens, forms, tables, dashboards.

Why this one. Screens are drawn FROM SETTINGS rather than written one by one. A tenant that renames a field, adds a column or turns a module off gets a different screen with no new code written — which is the only way one system can serve a steel plant and a single creator without becoming two systems.

What can replace it

- Vue or Svelte — the screen definitions are plain data and do not care what draws them
- Server-rendered pages where speed on a weak connection matters more than interaction
- A native mobile shell reading the same screen definitions

The rest of the code only ever talks to the screen definition format — so changing the line above changes one file, not the application.

What the move actually costs. Moderate, and bounded: what a screen contains is data, so a rewrite replaces the painter, not the paintings.

Background work — A queue backed by the database, with named workers

What it does. Does the things nobody should have to wait for — sending a thousand messages, building a month-end report, pulling orders overnight.

Why this one. Every job is written so that running it twice does the same thing as running it once. That single discipline is what makes it safe to retry after a failure, and it is worth more than any particular queue product.

What can replace it

- A Redis-backed queue when volume outgrows the database
- A hosted queue service, behind the same interface
- An external workflow tool such as n8n for steps a non-programmer should be able to edit

The rest of the code only ever talks to `JobQueue` — so changing the line above changes one file, not the application.

What the move actually costs. Low. Jobs are plain functions with a name; the queue only decides when they run.

Search — PostgreSQL full-text search

What it does. Finds a product, a customer or a document by a few typed letters, instantly.

Why this one. Postgres can search well enough for a long time, and starting there means one less thing running, one less thing to back up, and one less thing to keep in step with the database.

What can replace it

- OpenSearch or Elasticsearch when catalogues grow large
- Meilisearch or Typesense — small, fast, self-hostable
- A hosted search service behind the same interface

The rest of the code only ever talks to `SearchService` — so changing the line above changes one file, not the application.

What the move actually costs. Low, and it is a one-way door you can walk back through: the records stay in the database either way, so a search engine is only ever a faster copy.

Sign-in and permissions — Sessions issued by the platform, with permissions checked in the backend and again in the database

What it does. Proves who somebody is, then decides what they are allowed to see and change.

Why this one. Who you are and what you may do are kept apart deliberately. Sign-in can be handed to an outside service — or to a customer's own company login — while permissions stay ours, because they depend on the company and role structure no outside service knows about.

What can replace it

- An identity provider for sign-in only, with permissions still decided here
- A customer's own company sign-in, for enterprises that require it
- Self-hosted Keycloak or Authentik, when nothing may leave the building

The rest of the code only ever talks to `IdentityService` — so changing the line above changes one file, not the application.

What the move actually costs. Low for sign-in, by design. Permissions never move, so the expensive half is never in play.

Keys and passwords the system uses — Environment variables on the server, readable only by the service account

What it does. Holds the connection details and keys the software needs, away from the code.

Why this one. A key in the code is a key in every copy of the code forever. Keeping them outside means one can be replaced in a minute without changing a line.

What can replace it

- A managed secrets service, when there are enough of them to be worth it
- Self-hosted Vault or Infisical
- Encrypted files kept outside source control

The rest of the code only ever talks to `ConfigService` — so changing the line above changes one file, not the application.

What the move actually costs. Very low — the code asks for a name and does not care where the value came from.

Messages to customers and staff — A message service with one adapter per provider, per tenant

What it does. Sends WhatsApp messages, text messages and email — reminders, confirmations, statements.

Why this one. Each tenant connects its own accounts. The platform is built with a place for them to plug in and never holds one central account of its own — a business's conversations with its own customers belong to that business. The platform's job is the plug, not the account.

What can replace it

- Any WhatsApp provider — the adapter changes, the code that decides what to send does not
- Text message and email as fallbacks when a message cannot be delivered
- A shared inbox or an export, for a tenant with no messaging account at all

The rest of the code only ever talks to `MessageService` — so changing the line above changes one file, not the application.

What the move actually costs. One adapter per provider. Switching is a settings change made by the tenant, not a release made by us.

Storefronts and marketplaces — A channel adapter per storefront or marketplace

What it does. Brings orders in from a shop website or a marketplace, and sends stock and prices back out.

Why this one. Every one of these is treated as a channel with an adapter. Adding a marketplace is writing one adapter and creating one record — never a change to how orders work.

What can replace it

- A different storefront platform — a new adapter, and orders keep arriving
- File import for a channel with no connection available

- Manual entry, which must always remain possible

The rest of the code only ever talks to `ChannelAdapter` — so changing the line above changes one file, not the application.

What the move actually costs. One adapter each. The order, the stock number and the books never change shape.

Taking payments — A payment adapter per provider, with the card field hosted by the provider

What it does. Collects money from customers online.

Why this one. Card details are handed straight to the payment provider's own secured field and never touch this system — so there is nothing sensitive here to protect, and switching provider moves no card data, because none was ever held.

What can replace it

- Any other payment provider, behind the same interface
- Bank transfer and UPI details recorded against the invoice
- Cash on delivery, reconciled when the courier settles

The rest of the code only ever talks to `PaymentService` — so changing the line above changes one file, not the application.

What the move actually costs. One adapter. No card data ever moves, because none is ever stored.

Delivery and couriers — A courier adapter per carrier

What it does. Books a shipment, prints the label, and follows it to the door.

Why this one. Rate cards and tracking differ per courier; what a shipment IS does not.

What can replace it

- A courier aggregator, which is itself just one more adapter
- A different carrier directly
- Manual booking with the tracking number typed in — always available

The rest of the code only ever talks to `CourierService` — so changing the line above changes one file, not the application.

What the move actually costs. One adapter each.

Artificial intelligence — A router in front of several providers, ending on one that needs nothing bought

What it does. Writes descriptions, tags photographs, summarises, and answers questions about your own data.

Why this one. Ordered fallback, a breaker on anything failing repeatedly, and a spend ceiling that REFUSES rather than warns. Because every capability also has an option that costs nothing, a spent budget can stop the

spending without ever stopping the business.

What can replace it

- Any hosted model provider — an entry in the router, not a change to the system
- A model running on your own machine, for work that is routine or private
- Templates and rules with no model at all, which must always remain the last resort

The rest of the code only ever talks to `ModelRouter` — so changing the line above changes one file, not the application.

What the move actually costs. A list entry. The router exists precisely so changing provider is never a project.

Where it runs — Containers on a virtual server

What it does. The machines that serve the website and the application.

Why this one. The application is packaged as an ordinary container with nothing host-specific inside it. That single decision is what keeps every hosting option open, forever.

What can replace it

- A managed container platform, when scaling by hand stops being fun
- A different cloud, or a different country, for the same container
- A machine in your own office, for data that must not leave it

The rest of the code only ever talks to `the container image` — so changing the line above changes one file, not the application.

What the move actually costs. Low by construction. If moving hosts is ever hard, something host-specific has leaked in and that is the bug.

Source control and automatic checks — Git, with automatic checks on every change

What it does. Keeps the history of every change and runs every test before anything goes live.

Why this one. Git itself is the thing that matters, and git is not owned by anybody. The host is a convenience.

What can replace it

- A different hosting service — a git repository moves with one command
- Self-hosted Gitea or Forgejo
- A separate build service reading the same repository

The rest of the code only ever talks to `the test commands themselves` — so changing the line above changes one file, not the application.

What the move actually costs. Very low. The checks are ordinary commands, so any system that can run a command can run them.

Watching it — Structured logs and error reporting, in an open format

What it does. Reports errors, measures speed, and tells you when something stops answering.

Why this one. Standard formats mean the tool that reads them is replaceable without changing what the system emits.

What can replace it

- Any hosted error-tracking service
- Self-hosted GlitchTip, or a Grafana and Prometheus stack
- Log files plus an uptime checker, which is enough at the start

The rest of the code only ever talks to `Logger and the metric format` — so changing the line above changes one file, not the application.

What the move actually costs. Low — the system emits a standard shape and does not know who is reading it.

Making documents — HTML templates printed to PDF by a headless browser

What it does. Produces invoices, statements, labels and reports as files a person can print or send.

Why this one. What a document SAYS is data. How it is drawn is replaceable, and should be.

What can replace it

- A dedicated PDF library for very high volume
- A hosted document service
- Spreadsheet or CSV output, which some readers prefer anyway

The rest of the code only ever talks to `DocumentRenderer` — so changing the line above changes one file, not the application.

What the move actually costs. Low. Templates are content; the renderer is a tool.

M17 · WHAT A BUSINESS CAN CHANGE ITSELF

Rule 2 in full. Everything below is changed by the business, in the app, taking effect the same minute — no developer, no release, no phone call. And every change carries the date it starts from and is added rather than written over, so what was already recorded does not move.

18 things you can change, across 4 areas — and 6 that can never be switched off. Everything below is changed in the app, by you, taking effect the same minute. None of it needs a developer, a release or a phone call.

The column that matters most is the last one: **what happens to records already made.** A change carries the date it starts from and is added rather than written over, so a supervisor can leave on Tuesday and a replacement start

on Wednesday — and last month's payroll, already paid, does not move by a rupee. *Purana record mitta nahin; naye date se naya rule lagta hai.*

People

What changes	Who can	Takes effect	Records already made
Somebody joins — a worker, a supervisor, an office staff member, a contractor	Admin, or an HR role	They exist from their start date and can be assigned work the same minute.	Nothing before their start date mentions them, because they were not there.
Somebody leaves, with or without notice	Admin, or an HR role	Marked as left from a date. Their sign-in stops, and no new work is assigned to them. Their record is kept, not deleted.	Every hour they worked, every piece they made and every rupee they were paid stays exactly as it was. Deleting the person would blank all of it and change months already closed — so the record remains and simply has an end date.
A replacement starts immediately, in the same position	Admin, or an HR role	Added with their own start date and given the position. Work in progress is reassigned to them from that date. No waiting, no release, no developer.	Work completed under the previous person stays credited to the previous person. Two people held the same position at different times, and every record knows which one applied when.
A pay rate, a piece rate or a salary changes	Admin, or an HR role	Applies from the date you set — which may be today, a future date, or a past one you are correcting.	Every completed period recalculates to the rate that applied then, not the new one. This is the single most important line in this register: a rate that silently applied backwards would change payments already made to real people.
What somebody is allowed to see and do	Admin	Takes effect on their next action. Screens they may no longer open stop opening.	Everything they did while they held the old permissions stays recorded, with the permissions they had at the time.

Structure

What changes	Who can	Takes effect	Records already made
A new company is opened, or an existing one is closed	Admin	It exists immediately with its own name, trading name, code and document numbering. The group view includes it from that date.	Group figures for earlier periods are unchanged, because the company did not exist in them.
A new way of selling is added — a marketplace, a shop, a counter, an export desk	Admin	Orders can arrive through it the same day. It appears in every report that breaks figures down by channel.	Earlier reports keep their own columns. A channel that did not exist then does not appear then.
A godown, a shop, a unit or a stock point is added, renamed or closed	Admin	Stock can move to and from it immediately.	Stock movements already recorded keep pointing at it, under the name it had at the time.
Which parts of the system this business uses at all	Admin	Turned on, a module appears in the menu with its screens ready. Turned off, it disappears from the menu. A steel plant, a clothing brand and a single creator each end up with a different system built from identical code.	Turning a module off hides its screens and keeps its records. Nothing is destroyed by tidying a menu.

Your words

What changes	Who can	Takes effect	Records already made
What the system calls things	Admin	Every screen, every report and every document changes wording at once. One business says order, another says job, another says matter, consignment, batch or booking — the record underneath is identical.	Documents already issued keep the wording they were issued with, because that is what the customer received.
Extra information you want to record that nobody else needs	Admin	Added to the screen immediately, with the type you choose — text, number, date, a list to pick from, a yes or no. Reportable from the moment it exists.	Older records simply have no value for it, which is the truth. They are never back-filled with a guess.
The steps your work moves through	Admin	Add a stage, rename one, reorder them or remove one. New work follows the new list from that moment.	Work already part-way through keeps the stage it is in, even if that stage has since been removed — a job does not teleport because somebody edited a list.
The layout and numbering of invoices, statements, labels and reports	Admin	The next document uses the new layout or the new numbering.	Documents already issued are never re-rendered. What the customer holds and what you hold stay identical.

Rules

What changes	Who can	Takes effect	Records already made
Turning a discretionary rule on or off	Admin	Applies to the next transaction. One business requires an approval below a price floor; another does not — same software, different setting.	Transactions already posted are not re-judged against a rule that did not apply to them.
Who has to approve what, and above which amount	Admin	The next request follows the new path.	Requests already approved keep the path they went through, and the names of who approved them.
Tax rates and the categories they attach to	Admin, or an accounts role	Applies from its effective date, which for tax is set by law rather than by you.	Every invoice keeps the rate that applied on its own date. A return filed for an earlier period recalculates to that period's rate — this is not a convenience, it is the only correct behaviour.
Which outside service is used for messages, payments, delivery or artificial intelligence	Admin	The next message, payment or shipment goes through the new one.	Everything already sent keeps the record of which service carried it, which is what you need when you query one.
The most the system may spend on paid outside services	Admin	Enforced immediately. Over the ceiling, the paid option is refused and the work completes on one that costs nothing.	Spending already recorded is unchanged.

What can never be switched off

Short on purpose. Every line is something a bank, an auditor, a customer or an employee relies on, and a setting that could remove it would remove their protection too.

Never changeable	Why
The audit trail	Who changed what, and when. A system where this can be switched off cannot be used to answer a dispute, so it cannot be switched off.
Every record naming the company it belongs to	Without it, figures from two companies merge and no report can be trusted again.
One business being unable to read another's records	This is not a preference. It is the promise that makes a shared platform usable at all.
Money kept as exact whole units	The alternative loses fractions of a rupee in ways nobody can trace afterwards.
Deleting nothing — records are ended, never erased	An erased record changes a period that was already closed, filed and possibly audited.
Never asking for a marketplace, bank or account password	The system connects through proper keys that you can withdraw. A password would hand over an account you cannot take back.

M18 · EVERY TECHNICAL WORD, IN PLAIN LANGUAGE

No prior knowledge is assumed anywhere in this document. Every technical term it uses is here, in plain language, with an everyday comparison where one helps.

40 words. Every technical term this document uses, in plain language, with an everyday comparison. Nothing here assumes you already know any of them.

platform

One piece of software that many separate businesses use at the same time, each seeing only its own information.

Ek badi building jisme bahut saare offices hain. Building ek hai, par har office ki chaabi alag — koi kisi aur ke office mein nahin ghus sakta.

tenant

One business using the platform. Its people, its data and its settings are its own.

Us building mein ek office. Aapka office, aapka saamaan, aapka taala.

module

One area of work in the system — sales, purchase, staff, accounts. Each is a set of screens that belong together.

Dukaan ke alag-alag counters. Ek counter bikri ka, ek kharidi ka, ek hisaab-kitaab ka.

industry pack

A settings file that teaches the system your trade — what you call things, the stages your work moves through, the documents you issue.

Ek hi machine, alag-alag saancha. Saancha badal do, wahi machine doosri cheez banane lagti hai.

database

Where all the information is kept, arranged so any of it can be found instantly and nothing gets lost.

Ek badi almari jisme har cheez apne fix khaane mein rakhi hai — dhoondhne ke liye poori almari palatni nahin padti.

table

One kind of information inside the database — all your customers in one, all your orders in another.

Almari ka ek khaana. Ek khaane mein sirf customers, doosre mein sirf orders.

row

One single record — one customer, one order, one payment.

Register mein ek line. Ek line matlab ek entry.

schema

The written plan of what information the system keeps and how the pieces connect.

Makaan ka naksha. Deewar uthane se pehle kaagaz pe tay hota hai kaunsa kamra kahaan hai.

row-level security

A lock inside the database itself, so one business physically cannot read another business's records — even if the software above it has a bug.

Taala darwaze pe nahin, tijori pe. Guard so bhi jaaye toh bhi tijori band rehti hai.

migration

A recorded change to the shape of the database, so every copy of the system can be updated the same way, in the same order.

Naksha badla toh likh ke rakha — taaki har site pe wahi badlav, usi tarike se ho.

backup

A copy of everything, kept somewhere else, so a mistake or a failure does not lose your work.

Zaroori kaagzaat ki photocopy, doosri jagah rakhi hui. Asli jal jaaye toh bhi kaam nahin rukta.

cutover

The moment the business stops using the old way of working and starts using the new one for real.

Woh din jab purana tarika band aur naya shuru — ab asli kaam nayi jagah pe hoga.

integer paise

Money stored as a whole number of paise instead of a decimal, so amounts are exact and rounding can never quietly lose a rupee.

Paisa hamesha poore paise mein ginte hain, aadha-adhoora kabhi nahin — isliye hisaab kabhi ek rupya idhar-udhar nahin hota.

effective date

The date a change starts applying from. Records made before it keep the old value; records after it use the new one.

Naya rate 1 tarikh se lagu. Purane mahine ka bill purane rate se hi banega — woh apne aap nahin badlega.

audit trail

An automatic record of every change — what changed, who changed it, and when.

Har entry ke saath naam aur time apne aap likha jaata hai. Baad mein koi bole "maine nahin kiya", toh register bata deta hai.

backend

The part of the software you never see, which does the actual work — checks the rules, saves the records, calculates the totals.

Hotel ka kitchen. Customer nahin dekhta, par khaana wahin banta hai.

frontend

The part you see and click — the screens, the buttons, the forms.

Hotel ka dining hall aur menu card. Jo aapke saamne hai.

API

The agreed way two pieces of software talk to each other, so one can ask the other for something and get a predictable answer.

Waiter. Aap kitchen mein nahin jaate — waiter ko order dete ho, wahi khaana le aata hai. Waiter badal jaaye toh bhi order dene ka tarika wahi rehta hai.

interface

A written promise about what a part of the system does, without saying which product does it — so the product underneath can be swapped without anything above noticing.

Bijli ka socket. Socket ka size fix hai; usme koi bhi company ka plug lag jaata hai.

adapter

A small piece of code that translates between the system and one outside service, so the rest of the system never has to know which service is in use.

Travel adapter. Andar ka appliance wahi, bas plug ko us desk ke socket ke hisaab se badal diya.

storage

Where files are kept — photographs, invoices, scanned documents. Different from the database, which keeps information rather than files.

Almari ke bagal wala godown. Register almari mein, par bade dabbe aur photo godown mein.

cache

A small, fast copy of information that was just looked up, kept ready in case it is asked for again.

Counter pe rakha hua sabse zyada bikne wala saamaan. Har baar godown tak jaana nahin padta.

queue

A waiting line for work that does not have to finish this second — sending a hundred messages, building a big report.

Darzi ki dukaan ka parchi system. Kaam parchi pe likh ke lag gaya line mein; customer khada intezaar nahin karta.

job

One piece of work taken off the queue and done in the background.

Line mein se uthayi gayi ek parchi, ab uska kaam ho raha hai.

search index

A prepared list that makes finding things fast, the way the index at the back of a book beats reading every page.

Kitaab ke peeche wali index. Poori kitaab padhne ki zaroorat nahin, seedha page number mil jaata hai.

environment

A separate running copy of the system — one for trying things, one that customers actually use.

Rehearsal aur asli show. Practice alag jagah, taaki galti sabke saamne na ho.

deployment

Putting a new version of the software in place so people start using it.

Nayi dukaan kholna ya purani ko naya roop dena — jab tak shutter nahin uthta, customer ko farq nahin padta.

continuous integration

A robot that checks every change automatically, before anyone can put it live.

Quality-check wala banda gate pe khada. Har maal nikalne se pehle usse guzarta hai.

rollback

Putting the previous working version back, quickly, when a new one turns out to be wrong.

Naya taala kharab nikla toh purana taala wapas laga do — do minute ka kaam.

observability

Being able to see what the system is doing and what went wrong, without guessing.

Dukaan mein CCTV aur register. Kuch gadbad ho toh dekh sakte ho ki hua kya, andaaza nahin lagana padta.

uptime

How much of the time the system is actually working and reachable.

Dukaan mahine mein kitne din khuli rahi. Band rahi toh customer wapas chala gaya.

model

The piece of artificial intelligence that reads or writes text, tags a photograph, or answers a question.

Ek bahut padha-likha assistant. Kaam accha karta hai, par har baat pe usse poochho toh kharcha aur waqt dono lagta hai.

provider

A company whose service the system uses — for messages, for payments, for artificial intelligence, for delivery.

Supplier. Ek supplier maal na de toh doosre se le lo — kaam nahin rukna chahiye.

fallback

The next option the system automatically moves to when the first one fails or is unavailable.

Bijli gayi toh inverter. Inverter gaya toh mombatti. Andhera kabhi nahin hota.

spend ceiling

A maximum amount the system is allowed to spend on paid services, after which it refuses to spend more instead of warning you.

Jeb mein utne hi paise leke nikle jitna kharch karna hai. Khatam matlab khatam — udhaar nahin.

circuit breaker

A switch that takes a repeatedly failing service out of use for a while, instead of retrying it endlessly and slowing everything down.

Ghar ka MCB. Baar-baar fault aa raha hai toh woh line hi kaat deta hai, poora ghar band nahin hota.

role

What a person is allowed to see and do — a manager sees more than a counter staff member.

Chaabi ka guccha. Manager ke paas zyada chaabiyaan, staff ke paas kam.

permission

One specific thing a role is allowed to do, like approving a discount or viewing salaries.

Guchhe ki ek chaabi. Ek chaabi ek darwaza.

authentication

Proving you are who you say you are, usually by signing in.

Gate pe pehchaan dikhana. "Main kaun hoon" wala sawaal.

encryption

Scrambling information so that even somebody who steals the file cannot read it.

Apni hi code-bhasha mein likhna. Chori bhi ho jaaye toh padha nahin jaata.

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